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**DESIGN AND IMPLEMENTATION OF A PATIENT
MONITORING SYSTEM**

Final Report Submitted to the Department of Computer Engineering, Faculty of Engineering and Technology, University of Buea, in Partial Fulfilment of the Requirements for the Award of the Bachelor of Engineering (B.Eng.) Degree in computer Engineering

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Department of Computer Engineering

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CERTIFICATION OF ORIGINALITY

We the undersigned, hereby certify that this dissertation entitled “**Design and Implementation of a Patient Monitoring System**” presented by **Abanda Sergio Abanda** , Matriculation number **FE22A131** has been carried out by him in the Department of Computer Engineering, Faculty of Engineering and Technology, University of Buea under the supervision of **Associate Professor Elie Fute T.**

This dissertation is authentic and represents the fruits of his own research and efforts.

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DEDICATION

This work is dedicated to my beloved mother, **Mrs. Abanda Quinta Adong**, whose Unwavering love, sacrifices, prayers, and encouragement have been the foundation of my Academic and personal journey. Her strength and support continue to inspire me to strive for Excellence.

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ABSTRACT

This project presents the design and implementation of **HealthGuard**, a secure web-based Patient Monitoring System developed to improve the management and sharing of patient medical information among patients, doctors, and hospital administrators. Unlike traditional patient monitoring systems that rely on wearable sensors or Internet of Things (IoT) devices, the proposed system enables patients to upload medical documents such as laboratory reports, prescriptions, referral letters, consultation reports, and medical certificates while allowing doctors to request secure access to these records with the patient's approval. The system also provides appointment scheduling, prescription management, notifications, and a healthcare-focused Artificial Intelligence (AI) assistant capable of summarizing medical documents and responding only to healthcare-related questions. The application was developed using Next.js, Express.js, TypeScript, Prisma ORM, PostgreSQL, and Google Gemini AI. The implemented system demonstrates how modern web technologies and artificial intelligence can be integrated to improve healthcare information management while ensuring security, privacy, and efficient collaboration between patients and healthcare professionals.

Keywords: Patient Monitoring System, Electronic Health Records, Artificial Intelligence, Healthcare Information System, Medical Document Management, Role-Based Access Control.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CSS	Cascading Style Sheets
CRUD	Create, Read, Update and Delete
EHR	Electronic Health Record
EMR	Electronic Medical Record
ERD	Entity Relationship Diagram
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IoT	Internet of Things
JSON	JavaScript Object Notation
JWT	JSON Web Token
ORM	Object Relational Mapper
OTP	One-Time Password
PDF	Portable Document Format
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface
UML	Unified Modeling Language
URL	Uniform Resource Locator
UX	User Experience

CHAPTER 1: GENERAL INTRODUCTION

1.1 Background and Context of the Study

Healthcare institutions are increasingly adopting digital technologies to improve patient care, medical record management, and communication between healthcare professionals and patients. However, many healthcare facilities, particularly in developing countries such as Cameroon, still depend on paper-based medical records and manual procedures for managing patient information. These traditional methods often result in misplaced records, delays in accessing patient information, poor communication between healthcare providers, and reduced efficiency in healthcare delivery.

Patient monitoring has evolved beyond the continuous measurement of physiological parameters to include the effective management of patient health information throughout the healthcare process. Modern patient monitoring systems emphasize secure storage of medical records, appointment management, controlled information sharing, and timely communication between patients and healthcare professionals. Integrating Artificial Intelligence (AI) into such systems further enhances healthcare delivery by assisting users in understanding medical information and summarizing complex medical documents.

This project therefore proposes the design and implementation of a secure web-based Patient Monitoring System that enables patients to upload medical documents, allows doctors to request authorized access to patient records, and provides hospital administrators with tools to manage healthcare personnel within their institutions. The system also incorporates an AI-powered healthcare assistant to improve access to medical information while maintaining patient privacy and security.

1.2 Problem Statement

Many healthcare institutions still experience challenges in managing patient medical records efficiently due to the continued use of manual record-keeping systems and fragmented digital solutions. Patients often find it difficult to securely store and share their medical documents with different healthcare providers, while doctors may experience delays in obtaining the necessary patient information required for effective clinical decision-making. Furthermore, many existing healthcare systems either lack intelligent support features or are too expensive and complex for small and medium-sized healthcare institutions.

There is therefore a need for a secure, affordable, and user-friendly patient monitoring system that enables patients to manage their medical documents, facilitates secure collaboration between patients and doctors, supports appointment and prescription management, and incorporates Artificial Intelligence to assist users in understanding medical information while maintaining strict privacy and access control.

1.3 Objectives of the Study

1.3.1 General Objective

The main objective of this study is to design and implement a secure web-based patient monitoring system for efficient management, sharing, and monitoring of patient medical information among patients, doctors, and hospital administrators.

1.3.2 Specific Objectives

The specific objectives of this study are:

- To design and implement a secure role-based patient monitoring system for patients, doctors, and hospital administrators.
- To develop modules for medical document management, appointment scheduling, prescription management, patient access requests, and notifications.
- To integrate a healthcare-focused Artificial Intelligence assistant capable of summarizing medical documents and responding to healthcare-related questions while maintaining patient privacy and security.

1.4 Proposed Methodology

The methodology adopted for this project involves the analysis, design, development, testing, and evaluation of a secure web-based patient monitoring system. The development process began with gathering system requirements through literature review and analyzing the challenges associated with traditional paper-based patient record management systems.

The system was designed using software engineering modeling techniques, including Unified Modeling Language (UML) diagrams such as use case diagrams, sequence diagrams, class diagrams, and entity relationship diagrams to define the system architecture and functional requirements.

The implementation phase involved the development of both frontend and backend components. The frontend provides responsive user interfaces for patients, doctors, and hospital

administrators, while the backend manages authentication, authorization, database operations, appointment management, prescription management, document handling, notifications, and Artificial Intelligence services.

Patient medical documents are securely stored and protected using encryption and role-based access control. Patients retain full control over their medical records by approving or rejecting doctor access requests. The system also integrates Artificial Intelligence to summarize medical documents, explain medical reports in simple language, and answer healthcare-related questions while restricting responses to the healthcare domain.

Finally, the completed system was tested to evaluate its functionality, security, usability, responsiveness, and overall performance.

1.5 Research Questions

The study seeks to answer the following research questions:

- How can patient medical documents be managed securely and efficiently using a web-based platform?
- How can patients securely share their medical records with authorized healthcare professionals?
- How can appointment scheduling, prescription management, and notifications improve healthcare service delivery?
- How can Artificial Intelligence assist in summarizing medical documents and providing healthcare-related assistance without replacing professional medical advice?
- How can a secure patient monitoring system improve healthcare information management in Cameroon?

1.6 Significance of the Study

This study contributes to improving healthcare information management through the application of modern web technologies and Artificial Intelligence. The proposed patient monitoring system reduces the limitations associated with paper-based medical record management by providing a secure and centralized platform for storing, sharing, and managing patient medical documents.

The system improves collaboration between patients and healthcare professionals by allowing patients to securely upload medical documents and control access to their health information. Features such as appointment scheduling, prescription management, notifications, and role-based access control contribute to more efficient healthcare service delivery.

Furthermore, the integration of Artificial Intelligence assists users by summarizing medical documents, explaining medical reports in simple language, and providing healthcare-related information while emphasizing that medical decision should always be made by qualified healthcare professionals.

This study also contributes to the field of Software Engineering by demonstrating the application of secure software development practices, modern web technologies, database management, and Artificial Intelligence in the development of healthcare information systems. It may further serve as a foundation for future research and development of more advanced digital healthcare solutions in Cameroon and other developing countries.

1.7 Scope of the Study

This study focuses on the design and implementation of a secure web-based patient monitoring system for managing and sharing patient medical information digitally. The system provides separate interfaces for patients, doctors, and hospital administrators.

The project includes:

- Secure upload and management of patient medical documents.
- User authentication and role-based access control.
- Patient-controlled approval of doctor access requests.
- Appointment scheduling and prescription management.
- AI-assisted medical document summarization and healthcare-related assistance.
- Notification management.
- User-friendly dashboards for patients, doctors, and hospital administrators.
- The implementation focuses entirely on software functionalities and does not include physical medical devices, wearable sensors, or Internet of Things (IoT) technologies.

1.8 Delimitation of the Study

This project is limited to the software design and implementation of a patient monitoring system. The system will not include direct integration with physical monitoring devices such as heart rate sensors or temperature sensors.

The AI functionalities implemented in this project will focus mainly on medical data summarization, keyword extraction, and basic health insight generation rather than advanced medical diagnosis. In addition, the block chain logging mechanism will be simulated using traditional database technologies instead of deploying a fully decentralized block chain network.

The project will also be tested within a limited environment and may require additional improvements before deployment in large-scale hospital environments.

1.9 Definition of Keywords and Terms

- **Patient Monitoring System**

A digital healthcare system designed to manage, monitor, and analyze patient medical information electronically.

- **Electronic Health Record (EHR)**

A digital version of a patient's medical history containing health information such as diagnoses, medications, laboratory results, and treatment records.

- **Artificial Intelligence (AI)**

The simulation of human intelligence processes by computer systems to perform tasks such as data analysis, prediction, and decision support.

- **Access Control**

Security mechanisms used to regulate and manage access to system resources and patient data.

- **OTP (One-Time Password)**

A temporary verification code used to authenticate and authorize user access to sensitive information.

- **Encryption**

The process of converting data into a secure format to prevent unauthorized access.

- **Healthcare Professional**

A medical practitioner such as a doctor or nurse authorized to access and manages patient information within the system.

- **Database**

An organized collection of data stored electronically for easy access, management, and retrieval.

1.10 Organization of the Dissertation

This dissertation is organized into five chapters.

Chapter One: presents the general introduction of the study, including the background and context, problem statement, objectives, methodology, significance, scope, delimitation, definition of terms, and organization of the dissertation.

Chapter Two: presents the literature review and discusses existing patient monitoring systems, related concepts, and previous works relevant to the study.

Chapter Three: focuses on the analysis and design of the proposed system. It presents the methodology, system requirements, architecture, diagrams, and algorithms used in the design process.

Chapter Four: presents the implementation and results of the system. It describes the tools and technologies used implementation procedures, system interfaces, testing processes, and evaluation results.

Chapter Five: concludes the dissertation by summarizing the findings of the study, presenting recommendations, discussing challenges encountered, and suggesting areas for future work.

1.11 Partial Conclusion

This chapter introduced the background of the study, highlighted the challenges associated with traditional patient information management systems, and presented the objectives, research methodology, significance, scope, and limitations of the study. It established the need for a secure web-based patient monitoring system capable of improving the management and sharing of patient medical information while integrating Artificial Intelligence to enhance healthcare services. The next chapter reviews existing literature and technologies related to patient monitoring systems, electronic health records, healthcare information systems, and Artificial Intelligence in healthcare.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Healthcare systems across the world have experienced significant transformation due to advancements in Information and Communication Technology (ICT). The adoption of digital healthcare solutions has improved the storage, management, accessibility, and security of patient medical information. Patient monitoring systems and Electronic Health Record (EHR) systems are increasingly becoming essential components of modern healthcare institutions because they enable healthcare professionals to manage patient data efficiently while improving healthcare delivery and decision-making.

The integration of technologies such as Artificial Intelligence (AI), cloud computing, blockchain, and mobile health applications has further improved the efficiency of healthcare systems by enabling intelligent healthcare analysis, secure data management, and remote patient monitoring. Despite these advancements, many healthcare institutions in developing countries such as Cameroon still rely heavily on paper-based systems for managing patient records, leading to several operational and security challenges.

This chapter reviews relevant literature related to patient monitoring systems, electronic health records, digital healthcare technologies, Artificial Intelligence in healthcare, healthcare data security, and existing healthcare systems related to the proposed study. The review also identifies limitations in existing systems and establishes the research gap that justifies the need for the proposed system.

2.2 Concept of Patient Monitoring Systems

A patient monitoring system is a healthcare information system designed to collect, manage, monitor, and analyze patient medical information electronically. These systems assist healthcare professionals in observing patient conditions, tracking medical histories, and making informed medical decisions.

Patient monitoring systems are widely used in hospitals, clinics, and healthcare institutions to improve patient care and operational efficiency. They help reduce paperwork, minimize human errors, and provide quicker access to medical information. Modern monitoring systems also

support remote healthcare services, enabling healthcare providers to monitor patient conditions from different locations.

According to the World Health Organization (WHO), digital health technologies have become important tools for strengthening healthcare systems, improving healthcare accessibility, and supporting informed medical decision-making. WHO further explains that digital technologies improve interoperability, data sharing, and healthcare coordination.

The transition from traditional paper-based systems to electronic healthcare systems has significantly improved the management of patient records. Electronic systems provide faster access to patient information, better coordination between healthcare professionals, and improved continuity of care.

2.3 Related Works

Several digital healthcare systems have been developed to improve patient monitoring, healthcare management, medical record storage, and communication between healthcare professionals and patients. These systems integrate various technologies such as Electronic Health Records (EHR), Artificial Intelligence, cloud computing, and secure authentication mechanisms to improve healthcare delivery.

Although these systems provide significant advantages, many still face challenges related to implementation cost, data security, interoperability, infrastructure requirements, and adaptability in developing countries. The following section reviews some existing systems related to the proposed study.

2.3.1 Epic Electronic Health Record System

Epic Systems Corporation

Epic is one of the most widely used Electronic Health Record (EHR) systems in hospitals and healthcare institutions worldwide. The system provides functionalities such as patient record management, appointment scheduling, laboratory management, billing, and healthcare analytics.

Epic allows healthcare professionals to access patient records electronically and supports communication between hospitals and healthcare providers. The system also provides patient portals that enable patients to view their health records remotely.

Despite its advantages, Epic are expensive to implement and maintain. The system also requires significant technical infrastructure and specialized training, making it difficult for many healthcare institutions in developing countries to adopt effectively.

2.3.2 Cerner Healthcare Management System

Oracle Health (Cerner)

Cerner is a healthcare information management system designed to improve healthcare operations and patient record management. The system provides functionalities such as electronic medical records, clinical decision support, laboratory management, and healthcare reporting.

Cerner improves healthcare coordination by enabling healthcare professionals to access and share patient information digitally. The system also integrates analytical tools that support healthcare decision-making processes.

However, Cerner systems are associated with high implementation and maintenance costs. The complexity of deployment and infrastructure requirements also limit their adoption in low-resource healthcare environments.

2.3.3 OpenMRS

OpenMRS Official Website

OpenMRS is an open-source medical record system developed primarily for healthcare institutions in developing countries. The system supports functionalities such as patient registration, medical record management, appointment scheduling, and reporting.

OpenMRS is widely adopted because it is flexible, customizable, and less expensive compared to commercial healthcare systems. It also supports integration with other healthcare technologies and can be adapted to different healthcare environments.

Despite its advantages, OpenMRS requires technical expertise for deployment and maintenance. Some healthcare institutions also face challenges related to data security configuration and system scalability.

2.3.4 Meditech Electronic Health Record System MEDITECH Official Website

Meditech is an Electronic Health Record system designed to improve patient data management and healthcare coordination. The system provides functionalities such as patient monitoring, prescription management, laboratory management, and clinical documentation.

Meditech supports digital healthcare workflows and helps healthcare professionals retrieve patient information quickly and efficiently. The system also supports interoperability between different healthcare departments.

However, Meditech systems may require expensive infrastructure and regular maintenance. Some healthcare institutions also face usability challenges due to system complexity.

2.3.5 AI-Based Healthcare Monitoring Systems

Artificial Intelligence-based healthcare systems integrate intelligent algorithms into healthcare platforms to support medical data analysis, predictive analytics, disease detection, and healthcare recommendations.

Recent studies indicate that AI technologies improve the efficiency and accuracy of healthcare systems by assisting healthcare professionals in analyzing patient information and supporting medical decision-making processes. AI-assisted systems are increasingly being used for health risk prediction, medical record summarization, and clinical recommendations.

According to recent healthcare technology studies, AI-powered healthcare systems contribute significantly to improving patient monitoring and healthcare efficiency.

Despite their benefits, AI-based healthcare systems still face challenges related to:

- Data privacy,
- Ethical concerns,
- Dependence on high-quality datasets,
- Bias in AI algorithms,
- Regulatory limitations.

2.3.6 Existing Hospital Management Systems in Cameroon

Several hospitals in Cameroon still rely heavily on manual or semi-digital systems for storing and managing patient records. In many healthcare institutions, patient information is recorded using paper files, handwritten reports, and basic computer systems.

Although some hospitals have introduced computerized healthcare systems, many of these systems remain limited in terms of security, interoperability, intelligent healthcare analysis, and centralized monitoring functionalities.

Challenges affecting healthcare digitalization in Cameroon include:

- Poor ICT infrastructure,
- Limited financial resources,
- Insufficient cyber security measures,
- Limited technical expertise,
- Inadequate internet connectivity.

These challenges continue to affect the efficiency and accessibility of healthcare services within the country.

2.4 Electronic Health Records (EHR)

Electronic Health Records (EHRs) are digital versions of patient medical records containing information such as diagnoses, prescriptions, laboratory results, treatment history, allergies, and medical notes. EHR systems allow healthcare professionals to access and update patient information electronically.

The introduction of EHR systems has transformed healthcare delivery by improving the efficiency of patient data management and reducing the limitations associated with paper-based records. EHR systems support healthcare professionals in making timely and accurate medical decisions while improving patient safety.

Research on Electronic Health Records indicates that digital healthcare systems improve patient monitoring, reduce administrative workload, and enhance healthcare coordination.

Despite these benefits, EHR systems also face several challenges including:

- Data privacy concerns,
- Cyber security threats,
- System interoperability issues,
- High implementation costs,
- Limited technological infrastructure in developing countries.

Recent studies have shown that healthcare institutions continue to face difficulties related to data security and unauthorized access to patient information. Reports on healthcare data breaches emphasize the importance of implementing strong security mechanisms such as encryption, authentication, and access control within healthcare systems.

2.5 Artificial Intelligence in Healthcare

Artificial Intelligence (AI) refers to the simulation of human intelligence by computer systems to perform tasks such as learning, prediction, reasoning, and decision-making. AI technologies are increasingly being integrated into healthcare systems to support diagnosis, medical analysis, predictive modeling, and patient monitoring.

In healthcare environments, AI can assist in:

- Medical data analysis,
- Disease prediction,
- Clinical decision support,
- Medical record summarization,
- Risk detection,
- Intelligent healthcare recommendations.

Recent research indicates that AI technologies are improving the accuracy and efficiency of healthcare systems. Studies have shown that AI-assisted systems can support healthcare professionals in analyzing patient data and improving medical decision-making processes.

Natural Language Processing (NLP), a branch of AI, is widely used in Electronic Health Record systems to analyze unstructured medical text and extract meaningful clinical information. NLP technologies can assist healthcare professionals in understanding medical records more efficiently and identifying important health patterns.

Although AI technologies provide numerous advantages, their adoption in healthcare systems also presents challenges such as:

- Data privacy concerns,
- Ethical considerations,
- Bias in AI algorithms,
- Dependence on high-quality datasets,
- Regulatory challenges.

2.6 Security and Privacy in Healthcare Systems

Healthcare systems manage highly sensitive patient information, making security and privacy essential requirements in digital healthcare applications. Unauthorized access to medical information can compromise patient confidentiality and negatively affect healthcare delivery.

Healthcare systems commonly implement several security mechanisms including:

- User authentication,
- Role-based access control,
- Encryption,
- Secure communication protocols,
- Audit logging,
- Multi-factor authentication.

Research has shown that healthcare systems remain major targets for cyber-attacks because of the high value of medical data. Several healthcare organizations have experienced data breaches and network disruptions affecting access to patient records and healthcare services.

The implementation of secure authentication mechanisms such as One-Time Passwords (OTP) and encrypted storage systems can significantly improve patient data security and privacy.

2.7 Blockchain Technology in Healthcare

Blockchain technology is a decentralized digital ledger used for secure and transparent data storage. In healthcare systems, blockchain can improve the security, integrity, and traceability of patient medical records.

Blockchain technologies provide several advantages including:

- Improved data security,
- Tamper-resistant records,
- Better data sharing,
- Enhanced transparency,
- Improved trust between healthcare stakeholders.

Recent research on blockchain-based healthcare systems suggests that blockchain technologies can improve interoperability and data security in Electronic Health Record systems.

However, blockchain implementation in healthcare systems also faces several limitations including:

- High computational requirements,
- Scalability challenges,
- Complexity of deployment,
- Integration difficulties with existing healthcare systems.

For this reason, many healthcare software solutions implement lightweight logging and secure database mechanisms instead of fully decentralized blockchain infrastructures.

2.8 Limitations of Existing Systems

Despite the progress achieved in digital healthcare technologies, existing patient monitoring systems still present several limitations:

- High implementation costs,
- Poor interoperability,
- Limited security mechanisms,
- Inadequate accessibility in developing countries,
- Complex deployment procedures,
- Dependence on stable internet infrastructure,
- Limited AI integration in low-resource environments.

Many existing systems are not fully adapted to the technological and infrastructural realities of healthcare institutions in Cameroon.

2.9 Research Gap

Several studies have focused on Electronic Health Record systems, AI-assisted healthcare technologies, and patient monitoring systems. However, many existing systems remain expensive, overly complex, or difficult to implement in developing countries.

Additionally, many healthcare institutions in Cameroon still rely heavily on manual patient management systems that lack secure digital monitoring capabilities. Existing systems often fail to provide a balance between:

- affordability,
- usability,
- security,
- intelligent healthcare support,
- And efficient patient monitoring.

This study therefore seeks to design and implement a secure and intelligent Patient Monitoring System capable of improving patient data management, healthcare accessibility, and healthcare decision-making while remaining practical and adaptable to healthcare institutions in Cameroon.

2.10 Partial Conclusion

This chapter reviewed literature related to patient monitoring systems, Electronic Health Records, Artificial Intelligence in healthcare, healthcare security, blockchain technologies, and existing healthcare systems related to the proposed study.

The review revealed that digital healthcare technologies have significantly improved healthcare delivery and patient data management. However, several challenges including poor security, high implementation costs, limited interoperability, and inadequate infrastructure still affect healthcare systems, especially in developing countries.

The literature also showed that integrating AI functionalities, secure authentication mechanisms, and efficient patient monitoring features can significantly improve healthcare systems. Based on the identified research gaps, this study proposes the design and implementation of a secure Patient Monitoring System capable of addressing some of the limitations associated with existing healthcare systems.

CHAPTER 3: ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the analysis and design of the proposed Patient Monitoring System. Following the problem identified in Chapter One and the literature reviewed in Chapter Two, this chapter focuses on transforming the identified requirements into a practical software solution. It describes the methodology adopted during the development process, the functional and non-functional requirements of the system, and the overall design of the proposed solution.

The chapter further presents the software architecture, database design, system workflows, and the algorithms that govern the major functionalities of the application. Unified Modeling Language (UML) diagrams are used to illustrate the structure and behavior of the system, providing a clear representation of how different users interact with the application and how the various software components collaborate to achieve the intended objectives.

The proposed system is designed as a secure web-based application that supports three categories of users: patients, doctors, and hospital administrators. The system enables patients to upload and manage medical documents, allows doctors to request authorized access to patient records, facilitates appointment scheduling and prescription management, and incorporates a healthcare-focused Artificial Intelligence assistant capable of summarizing medical documents and responding to healthcare-related questions. The design also emphasizes security, scalability, maintainability, and usability to ensure that the developed solution meets modern healthcare information management requirements.

3.2 Methodology

The development of the proposed Patient Monitoring System followed the **Agile Software Development Methodology**. Agile was selected because it supports iterative and incremental software development, allowing system requirements to be refined throughout the development process. This methodology encourages continuous testing, regular feedback, and flexibility, making it suitable for the development of modern web-based applications.

The project was developed through several iterative phases, beginning with requirements gathering and analysis, followed by system planning, system design, implementation, testing, and

evaluation. At each stage, improvements were made based on the identified requirements and the functionality of the developed modules. This iterative approach ensured that the final system met the needs of patients, doctors, and hospital administrators while maintaining security, usability, and scalability.

The methodology adopted for this study consisted of the following phases:

3.2.1. Requirements Analysis

This phase involved identifying and analyzing the functional and non-functional requirements of the proposed Patient Monitoring System. Information was gathered through literature review, observation of existing healthcare systems, and analysis of the challenges associated with traditional paper-based medical record management. The requirements of the three primary users: patients, doctors, and hospital administrators were carefully identified and documented.

3.2.2. System Design

During this phase, the overall structure of the application was planned. The database structure, system architecture, user interfaces, and interactions between different modules were designed before implementation. This stage provided a clear blueprint for the development of the system and ensured that all functional requirements were adequately addressed.

3.2.3. System Implementation

The implementation phase involved developing the frontend and backend components of the application. The frontend was developed using Next.js, React, TypeScript, and Tailwind CSS to provide a responsive and user-friendly interface. The backend was implemented using Express.js and TypeScript, while Prisma ORM and PostgreSQL were used for database management. Core functionalities such as authentication, document management, appointment scheduling, prescription management, notifications, and role-based access control were implemented during this phase. The system also integrated Google Gemini AI to summarize medical documents and provide healthcare-related assistance.

3.2.4. System Testing

After implementation, the system was tested to verify that each module functioned correctly and satisfied the specified requirements. Functional testing was carried out to ensure that features such as authentication, document upload, appointment management, access requests, prescription

management, notifications, and AI services operated as expected. Additional testing was performed to verify system security, responsiveness, and overall usability across different devices.

3.2.5. System Evaluation

The final phase involved evaluating the completed system against the objectives of the study. The evaluation focused on determining whether the developed application successfully addressed the identified healthcare information management challenges while providing a secure, efficient, and user-friendly platform for patients, doctors, and hospital administrators.

3.3 Design

The design phase translates the system requirements identified during the analysis phase into a structured software solution. It provides a blueprint for the implementation of the Patient Monitoring System by defining the system functionalities, user interactions, software components, and database structure. The design emphasizes security, usability, scalability, and maintainability to ensure that the developed application meets the needs of patients, doctors, and hospital administrators.

The following subsections present the functional and non-functional requirements of the proposed system together with the Unified Modeling Language (UML) diagrams used to model the behavior and structure of the application.

3.3.1 Functional Requirements

Functional requirements describe the services and operations that the Patient Monitoring System must perform in order to satisfy the needs of its users.

The system provides three categories of users, namely patients, doctors, and hospital administrators, each with different responsibilities and privileges.

Patient

The patient shall be able to:

- Register and log into the system.
- Manage personal profile information.
- Upload and manage medical documents.
- View uploaded medical documents.
- Approve or reject doctor access requests.
- Schedule appointments with doctors.

- View appointment history.
- View prescriptions issued by doctors.
- Receive system notifications.
- Interact with the AI healthcare assistant.

Doctor

The doctor shall be able to:

- Log into the system.
- Manage personal profile information.
- Search for patients.
- Request access to patient medical records.
- View authorized patient medical documents.
- Schedule and manage appointments.
- Create and manage patient prescriptions.
- Receive notifications.
- Use the AI assistant to summarize medical documents and obtain healthcare-related assistance.

Hospital Administrator

The hospital administrator shall be able to:

- Log into the system.
- Manage hospital information.
- Register, update, activate, deactivate, and remove doctors within the hospital.
- View doctors associated with the hospital.
- View patients associated with each doctor.
- Monitor doctor activities through notifications.
- Manage system users under the hospital.

3.3.2 Non-Functional Requirements

The proposed Patient Monitoring System satisfies the following non-functional requirements.

Security

The system protects sensitive medical information through authentication, role-based access control, encrypted password storage, and patient-controlled authorization of medical record access.

Performance

The system should provide fast response times during authentication, document retrieval, appointment management, and AI-assisted services.

Usability

The application provides a simple, responsive, and intuitive user interface that can be accessed using desktop computers, tablets, and mobile devices.

Reliability

The system is designed to ensure accurate processing of user requests while maintaining data integrity and minimizing system failures.

Scalability

The architecture allows the system to accommodate increasing numbers of users, hospitals, medical documents, and appointments without significant performance degradation.

Maintainability

The application follows a modular architecture that simplifies future maintenance, debugging, and the addition of new features.

3.3.3 Use Case Diagram

The Use Case Diagram illustrates the interactions between the different actors and the Patient Monitoring System. It provides a high-level representation of the functionalities available to each category of user.

The proposed system consists of three primary actors:

- Patient
- Doctor
- Hospital Administrator

Patients interact with the system by uploading medical documents, scheduling appointments, managing their profiles, viewing prescriptions, approving or rejecting doctor access requests, and communicating with the AI healthcare assistant.

Doctors use the system to manage appointments, request access to patient records, review authorized medical documents, create prescriptions, and utilize the AI assistant for healthcare-related support.

Hospital administrators are responsible for managing hospital information, registering and managing doctors, monitoring doctor activities, and maintaining the overall administration of the hospital within the system.

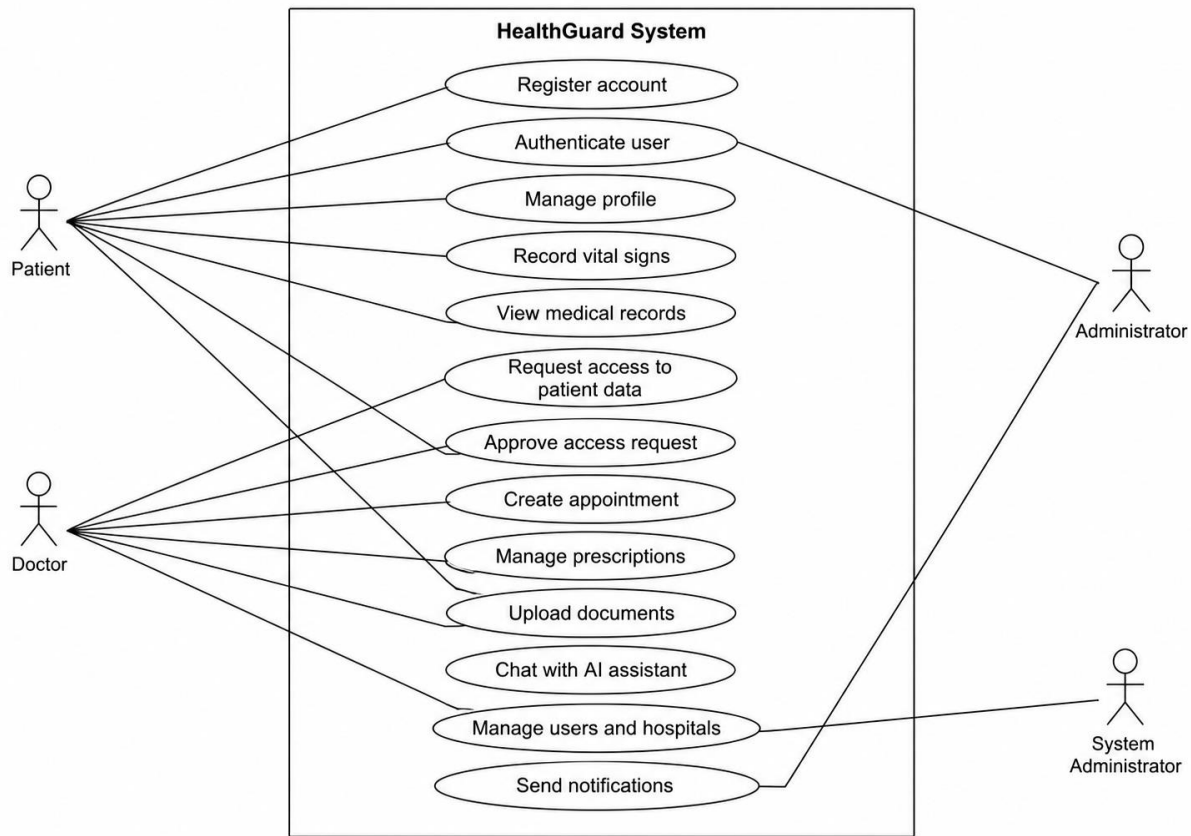


Figure 1: Use Case Diagram of the Proposed Patient Monitoring System

3.3.4 Class Diagram

The Class Diagram describes the static structure of the proposed Patient Monitoring System by illustrating the major classes, their attributes, operations, and relationships. It serves as the foundation for the object-oriented implementation of the application.

Some of the principal classes in the system include **Patient**, **Doctor**, **Hospital**, **Document**, **Appointment**, **Prescription**, **AccessRequest**, and **Notification**.

The **Patient** class stores personal information and maintains relationships with uploaded medical documents, appointments, prescriptions, and access requests.

The **Doctor** class contains professional information and manages appointments, prescriptions, authorized patients, and access requests.

The **Document** class stores information relating to uploaded healthcare documents such as laboratory reports, consultation reports, prescriptions, referral letters, and medical certificates.

The **Appointment** class records scheduled consultations between patients and doctors, while the **Prescription** class maintains medication information issued by doctors.

These classes collaborate to support secure healthcare information management and controlled sharing of patient medical records.

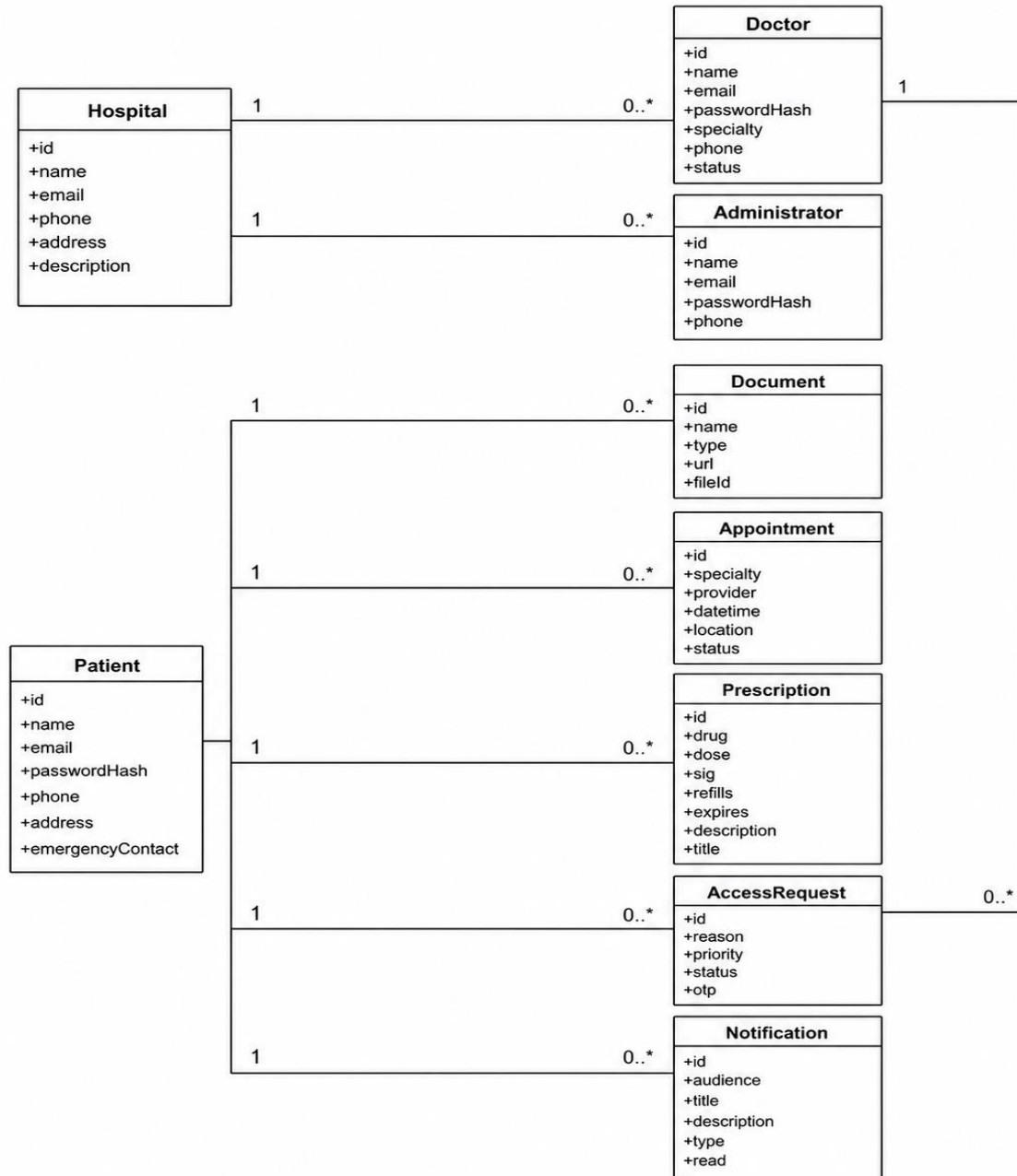


Figure 2: Class Diagram of the Proposed Patient Monitoring System

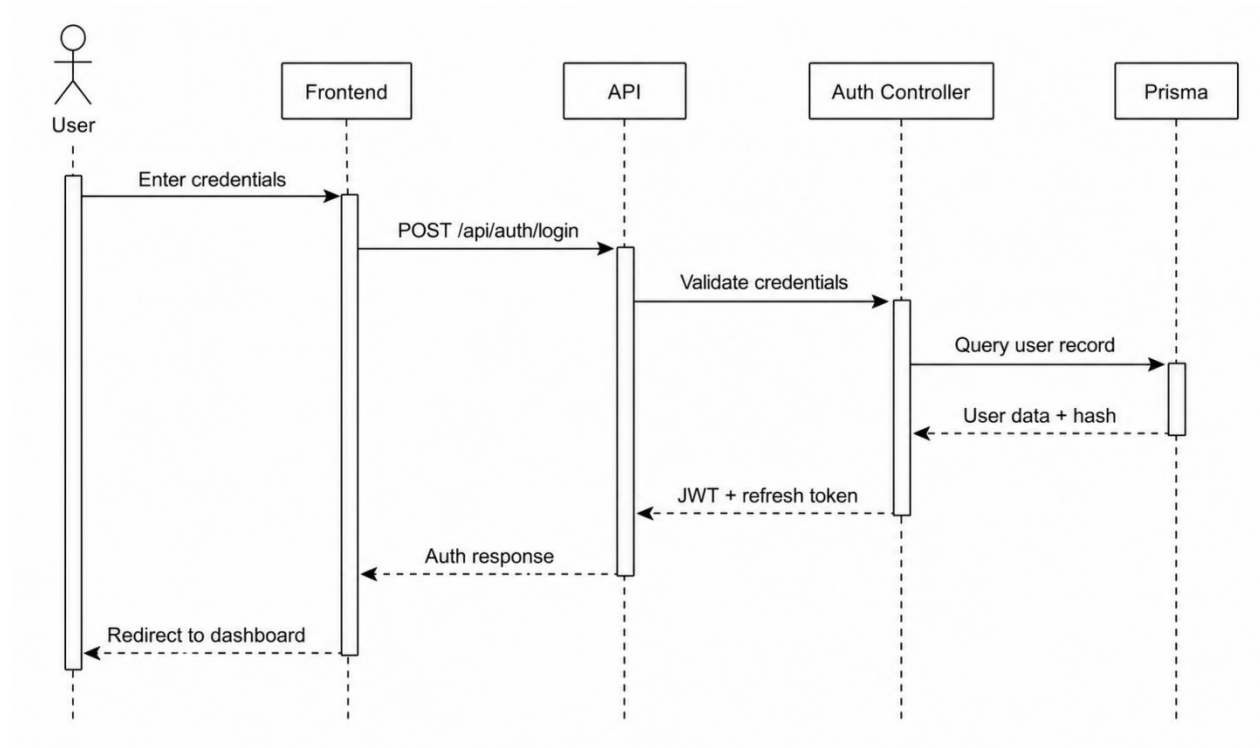
3.3.5 Sequence Diagram

The Sequence Diagram illustrates the chronological interaction between system components during the execution of a particular process. It demonstrates how messages are exchanged between users, the application, and the database to complete a specific operation.

One of the most important processes in the proposed system is the doctor's request for access to a patient's medical records.

The process begins when the doctor submits an access request through the application. The system records the request and sends a notification to the patient. The patient reviews the request and either approves or rejects it. If approved, the system grants the doctor permission to access the patient's medical documents. Otherwise, access is denied.

This sequence ensures that patients maintain full control over access to their medical information while enabling secure collaboration with healthcare professionals.



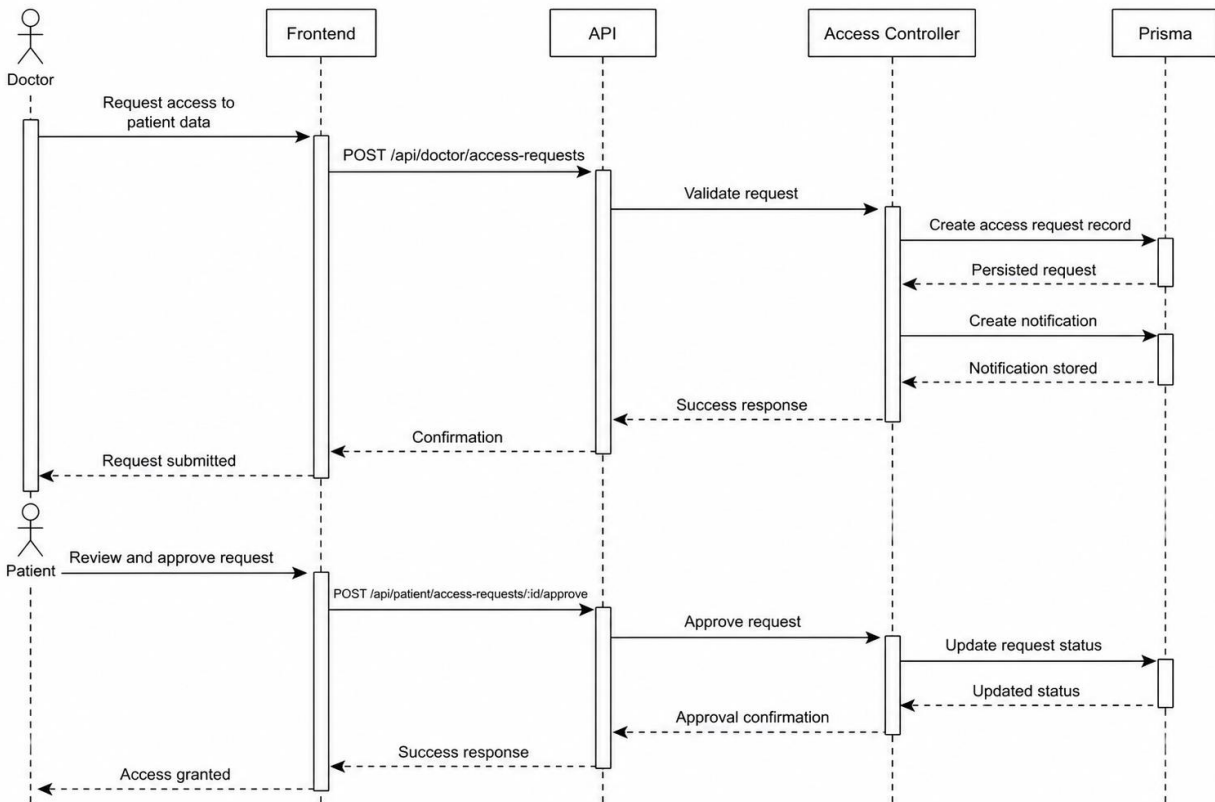


Figure 3: Sequence Diagram for Patient Access Request Process

3.3.6 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) illustrates the logical organization of the database and the relationships among the various entities used by the Patient Monitoring System.

The database consists of several entities including **Hospital**, **Administrator**, **Doctor**, **Patient**, **Document**, **Appointment**, **Prescription**, **AccessRequest**, and **Notification**.

A hospital can have many doctors, while each doctor belongs to one hospital. A patient can upload multiple medical documents, schedule several appointments, and receive multiple prescriptions. Doctors may create many prescriptions and manage multiple appointments. Access requests establish controlled relationships between doctors and patients, allowing doctors to view patient medical records only after receiving patient approval.

The ERD ensures data consistency by defining appropriate relationships, primary keys, and foreign keys between entities.

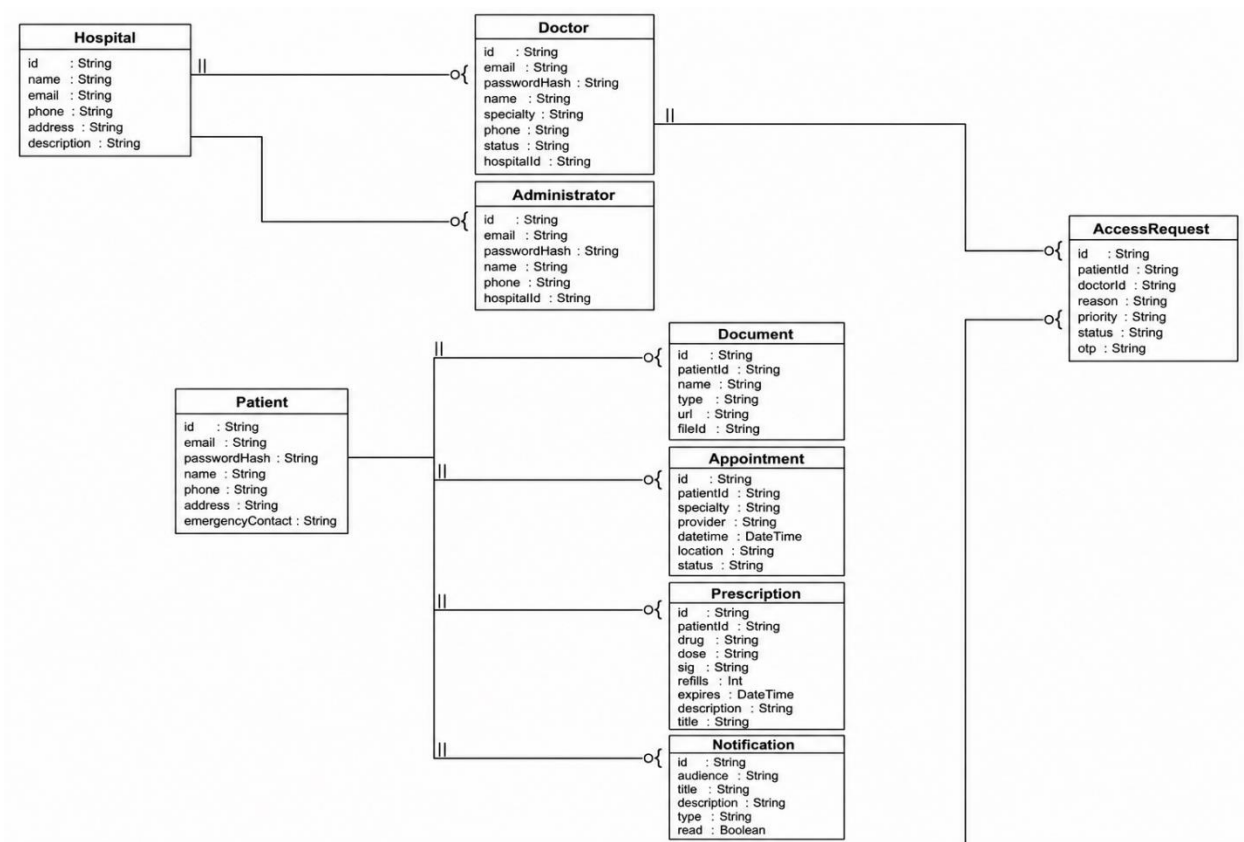


Figure 4: Entity Relationship Diagram of the Proposed Patient Monitoring System

3.4. Global Architecture of the solution

The proposed Patient Monitoring System adopts **three-tier architecture**, consisting of the presentation layer, application layer, and data layer. This architectural approach separates the user interface, business logic, and database management, thereby improving system maintainability, scalability, and security. The architecture also facilitates independent development and maintenance of each component while ensuring efficient communication between them.

The **presentation layer** represents the frontend of the application developed using Next.js, React, TypeScript, and Tailwind CSS. It provides responsive user interfaces through which patients, doctors, and hospital administrators interact with the system. This layer is responsible for displaying information, collecting user input, and communicating with the backend through RESTful API requests.

The **application layer** consists of the backend services developed using Express.js and TypeScript. This layer handles user authentication, authorization, appointment management, prescription management, document management, notifications, access requests, and

communication with the Artificial Intelligence service. It processes incoming requests from the frontend, applies the necessary business rules, and returns appropriate responses to users.

The **data layer** is implemented using PostgreSQL with Prisma ORM. It securely stores all system data, including user accounts, hospitals, medical documents, appointments, prescriptions, notifications, and access requests. Prisma ORM provides efficient interaction between the application and the database while ensuring data integrity and simplifying database operations.

The system also integrates the Google Gemini Artificial Intelligence service through secured API communication. The AI service is responsible for summarizing uploaded medical documents, explaining medical reports in simple language, and responding exclusively to healthcare-related questions while discouraging users from relying on AI for medical diagnosis or treatment decisions.

The interaction among these layers ensures that requests originating from users are securely processed before the required information is retrieved from or stored in the database and appropriate responses are returned to the user.

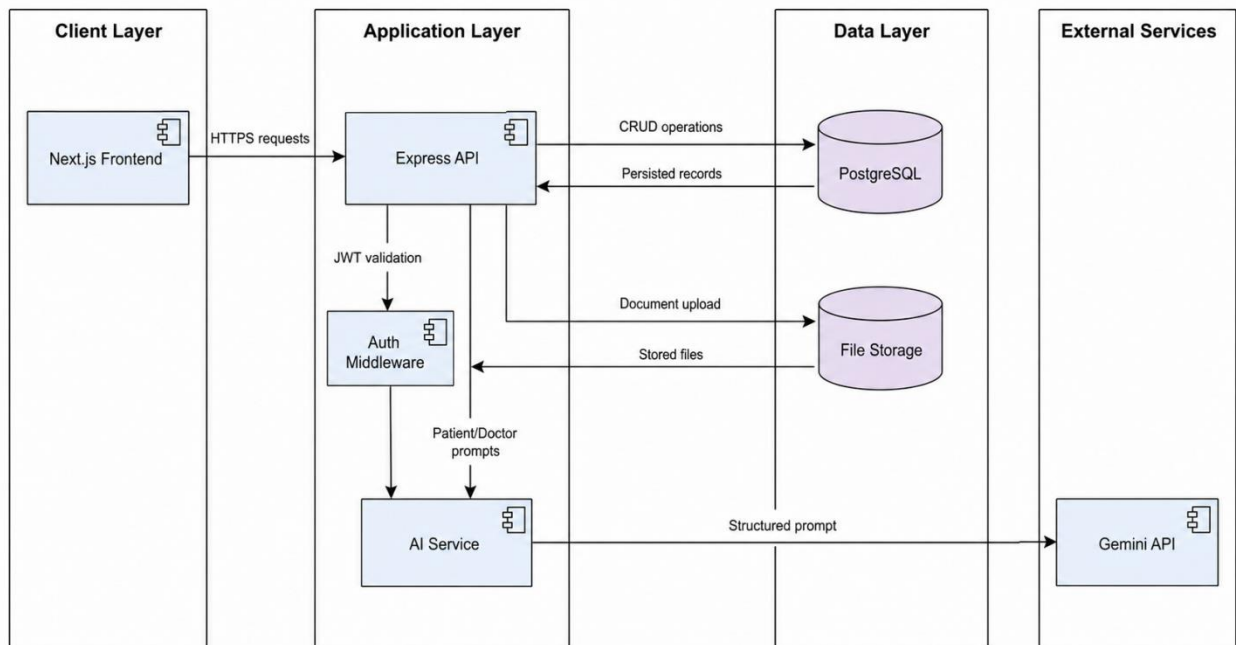


Figure 5: Global Architecture of the Patient Monitoring System

3.5 Description of the Algorithms

The Patient Monitoring System employs several algorithms to perform its core functionalities efficiently and securely. The following algorithms describe the logical processes implemented within the application.

Algorithm 1: User Authentication

This algorithm authenticates users based on their registered credentials and assigned roles.

1. User enters email and password.
2. System validates the input.
3. System searches for the user in the database.
4. Password is compared with the stored encrypted password.
5. If authentication succeeds, a JSON Web Token (JWT) is generated.
6. User is redirected to the appropriate dashboard according to their role.
7. Otherwise, an authentication error message is returned.

Algorithm 2: Medical Document Upload

This algorithm enables patients to securely upload medical documents.

1. Patient selects a medical document.
2. System validates the file type and size.
3. Document is uploaded and stored securely.
4. Metadata is saved in the database.
5. Success notification is returned to the patient.

Algorithm 3: Doctor Access Request

This algorithm manages patient-controlled access to medical records.

1. Doctor searches for a patient.
2. Doctor submits an access request.
3. System stores the request.
4. Patient receives a notification.
5. Patient approves or rejects the request.

6. If approved, doctor receives permission to access the patient's medical documents.
7. Otherwise, access is denied.

Algorithm 4: Appointment Scheduling

This algorithm manages appointment creation between patients and doctors.

1. Patient selects a doctor.
2. Patient provides appointment details.
3. System verifies the submitted information.
4. Appointment is stored in the database.
5. Doctor receives an appointment notification.
6. Appointment status is updated accordingly.

Algorithm 5: Prescription Management

This algorithm allows doctors to issue prescriptions after reviewing patient information.

1. Doctor selects a patient.
2. Doctor creates a prescription.
3. Prescription information is validated.
4. Prescription is saved in the database.
5. Patient receives a notification.
6. Prescription becomes available on the patient's dashboard.

Algorithm 6: AI Medical Document Summarization

This algorithm summarizes uploaded medical documents.

1. User selects a medical document.
2. Document content is extracted.
3. Extracted text is sent to the Google Gemini AI service.
4. AI generates a concise summary.
5. Summary is returned to the application.
6. User views the summarized information.

3.6 Description of the Resolution Process

The resolution process describes how the proposed Patient Monitoring System addresses the challenges identified during the problem analysis phase.

The process begins with user authentication, where patients, doctors, and hospital administrators securely access the system according to their assigned roles. Once authenticated, patients can upload medical documents, manage appointments, and control access to their medical information by approving or rejecting requests submitted by doctors.

Doctors interact with the system by requesting access to patient records, reviewing authorized medical documents, scheduling appointments, issuing prescriptions, and utilizing the integrated Artificial Intelligence assistant to summarize lengthy medical documents and obtain healthcare-related information.

Hospital administrators oversee hospital operations within the system by managing doctors, updating hospital information, activating or deactivating doctor accounts, and monitoring important activities through the notification system.

Throughout the entire process, the backend validates all requests, enforces role-based access control, processes business logic, and stores data securely within the PostgreSQL database.

Where Artificial Intelligence services are requested, the backend communicates with the Google Gemini API before returning summarized medical information or healthcare-related responses to the user.

This structured workflow ensures secure management of healthcare information while facilitating collaboration among patients, doctors, and hospital administrators.

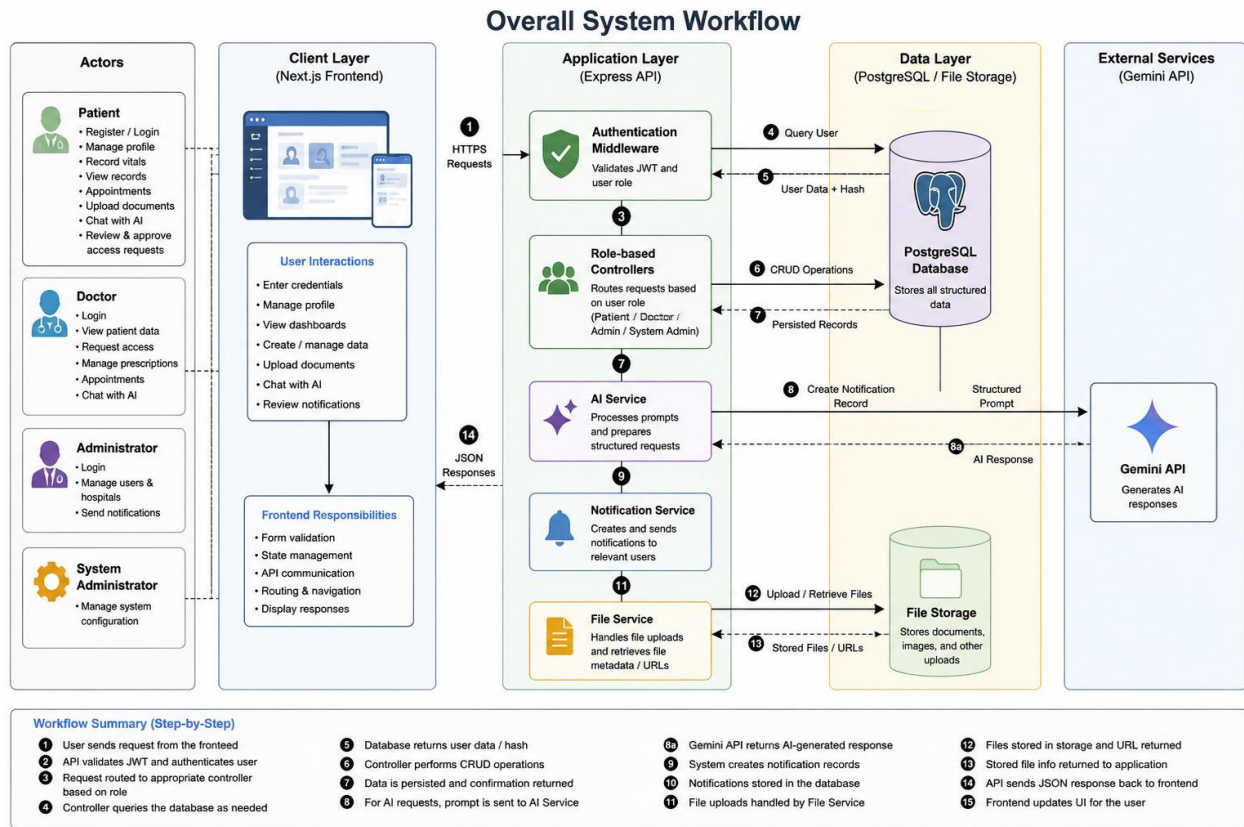


Figure 6: Overall System Workflow

3.7 Partial Conclusion

This chapter presented the analysis and design of the proposed Patient Monitoring System. It described the development methodology adopted for the project, the functional and non-functional requirements of the system, and the design models used to represent its structure and behavior. The chapter also discussed the global architecture of the solution, the algorithms governing the major system functionalities, and the overall resolution process adopted to address the identified healthcare information management challenges. The design provides a secure, scalable, and user-friendly foundation for the implementation of the system. The next chapter presents the implementation of the proposed solution, the user interfaces, and the testing procedures carried out to evaluate the performance and functionality of the developed application.

CHAPTER 4: IMPLEMENTAION AND RESULTS

4.1 Introduction

This chapter presents the implementation of the proposed Patient Monitoring System and the results obtained after its development. It describes the technologies and tools used during implementation, the development process followed in building the application, and the major functionalities that were successfully implemented. Screenshots of the developed system are presented to demonstrate the operation of the various modules available to patients, doctors, and hospital administrators.

The chapter further evaluates the performance of the developed system by comparing the implemented functionalities with the objectives stated in Chapter One. The evaluation demonstrates how the proposed solution improves healthcare information management through secure medical document management, controlled patient-doctor collaboration, appointment scheduling, prescription management, and Artificial Intelligence-assisted healthcare support.

4.2 Tools and Materials Used

The implementation of the proposed Patient Monitoring System was carried out using modern web development technologies selected for their scalability, reliability, maintainability, and security.

Tool/Technology	Purpose
React	User interface development
TypeScript	Type-safe application development
Next.js	Frontend development
Tailwind CSS	User interface styling
Express.js	Backend REST API development
Prisma ORM	Database access and management
PostgreSQL	Database management system
JSON Web Token (JWT)	Authentication and authorization
Google Gemini AI	AI-powered medical document summarization and healthcare assistance
Visual Studio Code	Source code development

Git & GitHub	Version control
Postman	API testing
Node.js	Runtime environment

Table 1: Tools and Materials Used

4.3 Description of the Implementation Process

The implementation of the Patient Monitoring System followed the Agile Software Development Methodology described in Chapter Three. Development was carried out incrementally, with each module implemented and tested before integrating it into the complete system.

The backend of the application was first developed by designing the database schema using Prisma ORM and PostgreSQL. Database models were created for hospitals, administrators, doctors, patients, medical documents, appointments, prescriptions, access requests, notifications, and user authentication.

After the backend APIs were completed, the frontend interfaces were developed using Next.js, React, TypeScript, and Tailwind CSS. Separate dashboards were implemented for patients, doctors, and hospital administrators based on their respective roles and privileges.

Authentication and role-based access control were implemented using JSON Web Tokens (JWT), ensuring that users could only access resources permitted by their assigned roles.

The Artificial Intelligence module was integrated using the Google Gemini API. The AI assistant was configured to summarize uploaded medical documents, explain medical reports in simple language, and respond exclusively to healthcare-related questions.

Finally, all system modules were integrated and tested to ensure proper communication between the frontend, backend, database, and AI services.

4.4 Presentation and Interpretation of Results

This section presents the major interfaces developed for the Patient Monitoring System together with explanations of their functionality.

4.4.1 Landing Page

The landing page serves as the entry point to the application. It introduces the HealthGuard platform, highlights its major features, and provides navigation links for user registration and login.

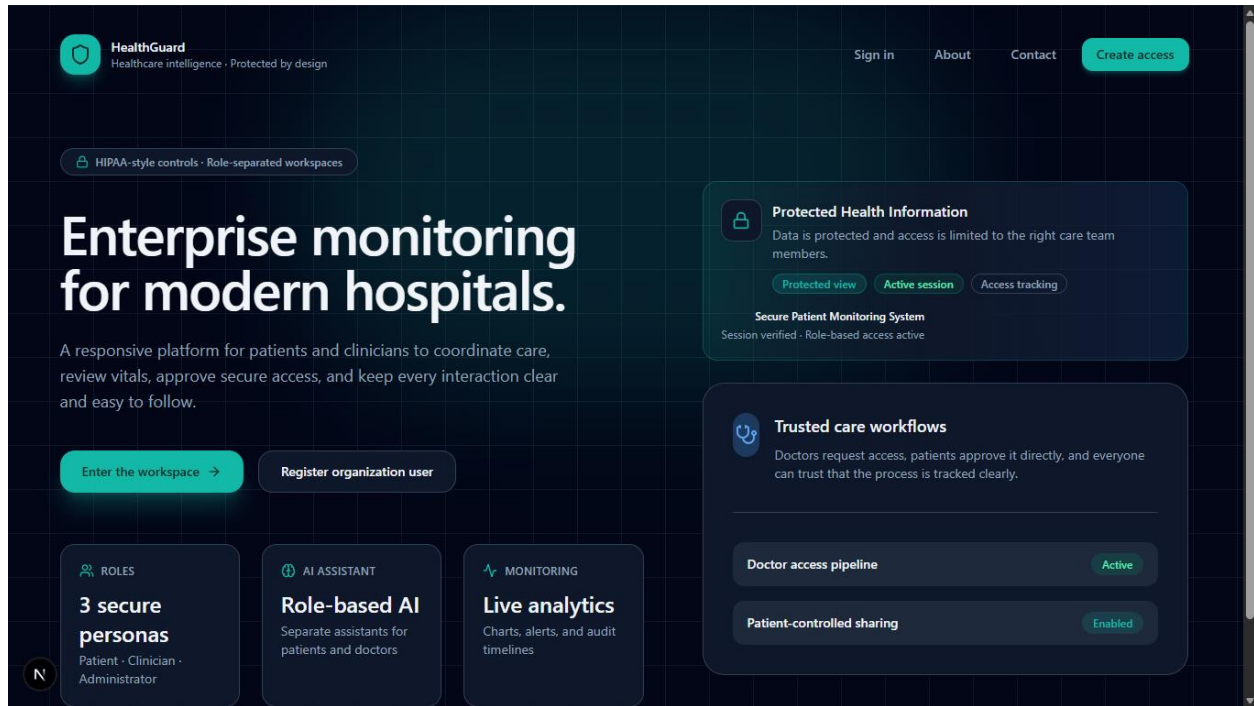


Figure 7: Landing Page

4.4.2 User Authentication

The authentication module enables patients, doctors, and hospital administrators to securely register and log into the system using their credentials.

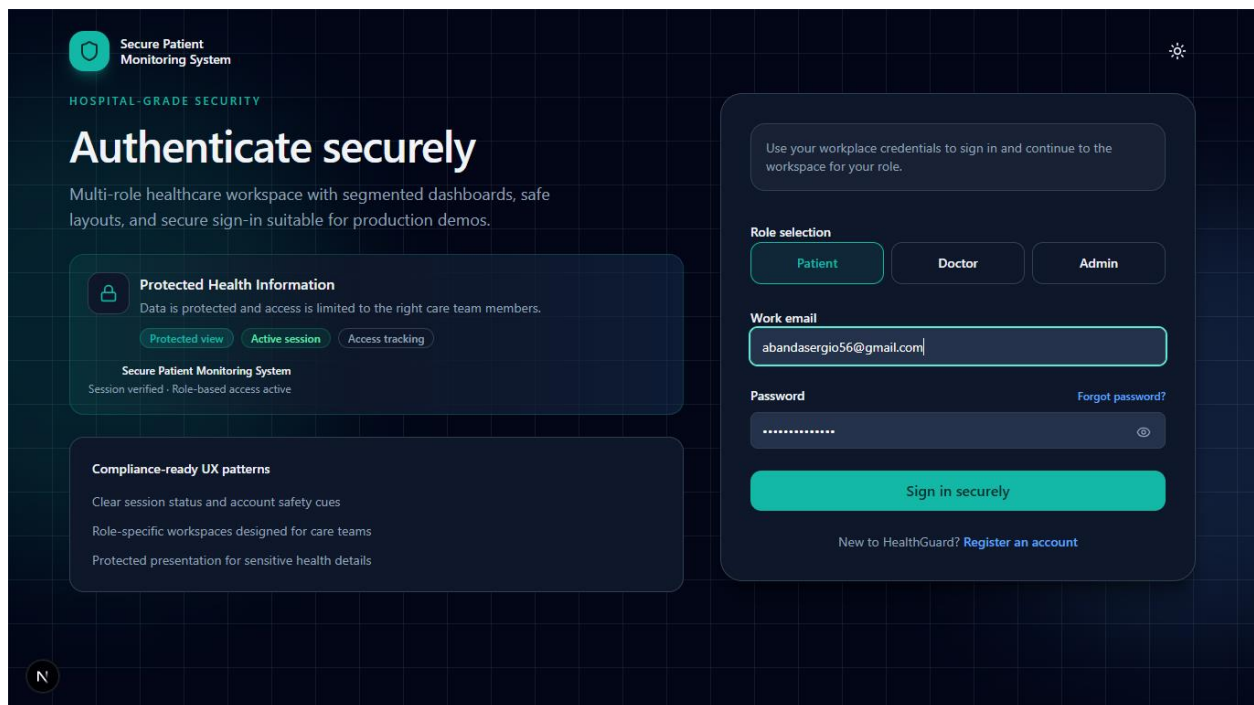


Figure 8: Login Page

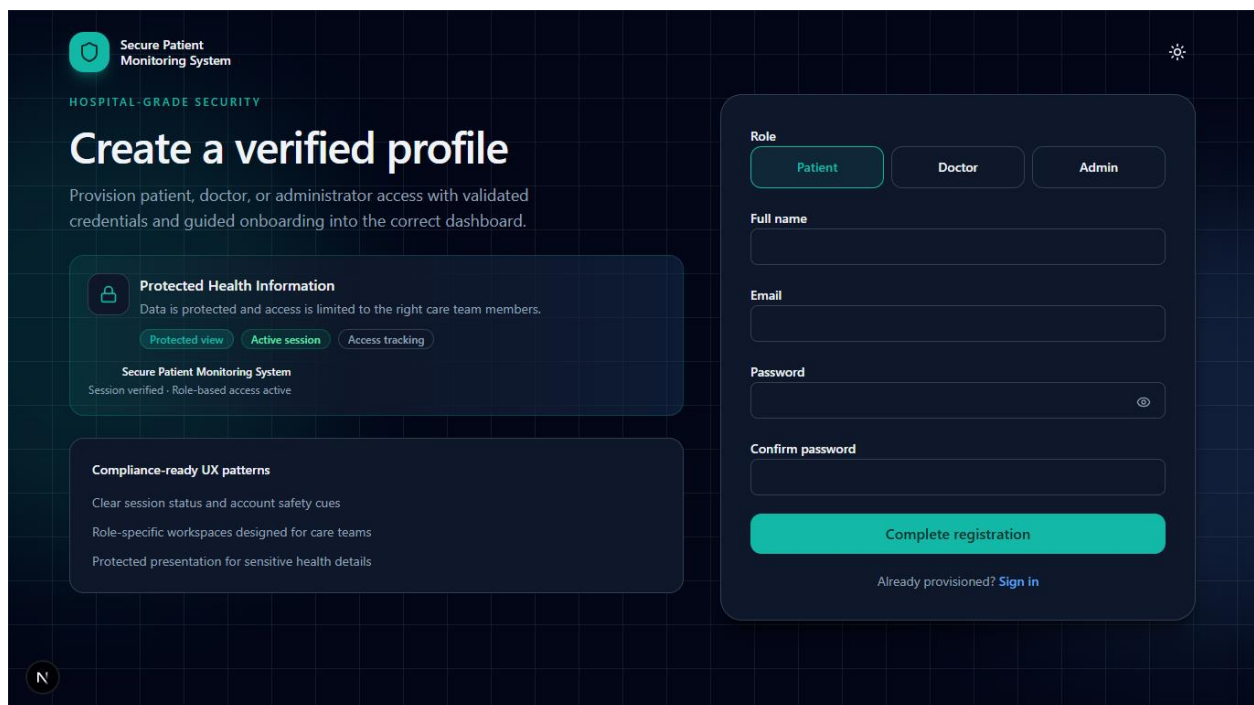


Figure 9: Registration Page

4.4.3 Patient Dashboard

The patient dashboard provides patients with a centralized interface for managing their healthcare information.

Patients can:

- Upload medical documents.
- View uploaded documents.
- Manage appointments.
- View prescriptions.
- Approve or reject doctor access requests.
- Receive notifications.
- Interact with the AI healthcare assistant.

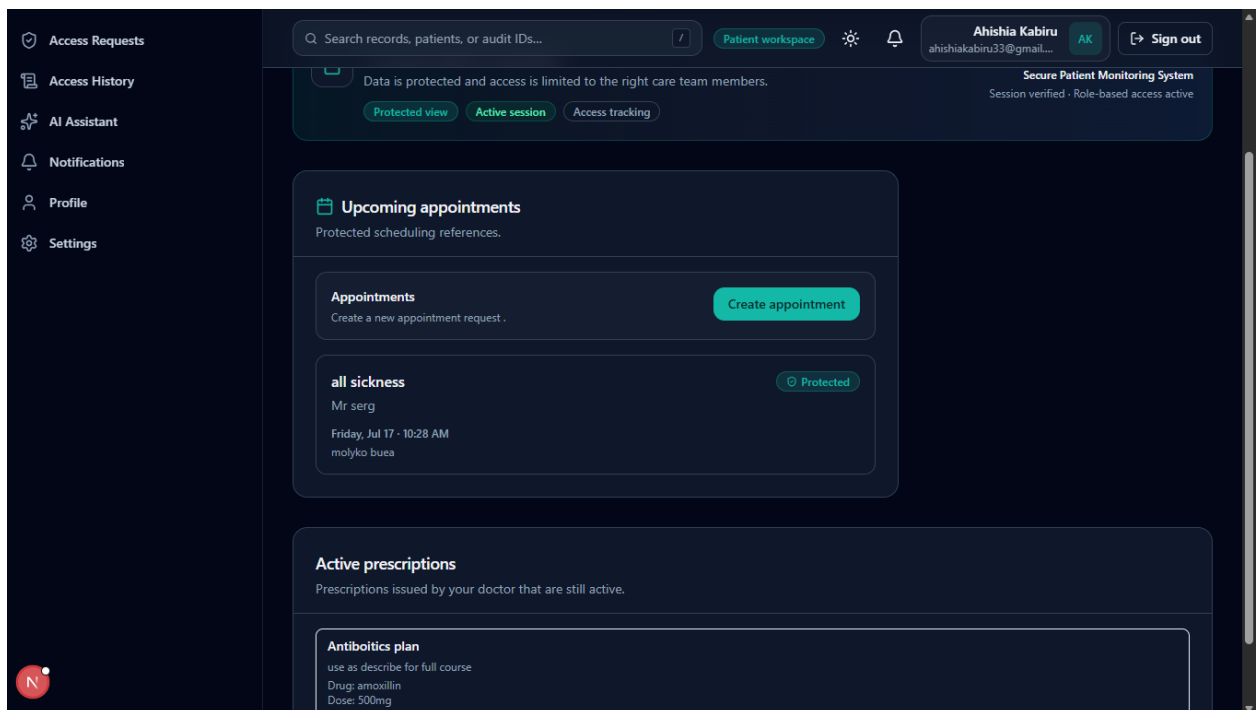


Figure 10: Patient Dashboard

4.4.4 Doctor Dashboard

The doctor dashboard enables healthcare professionals to manage patients assigned to them and monitor their daily activities.

Doctors can:

- Search for patients.
- Request access to patient records.
- Review authorized medical documents.
- Schedule appointments.
- Create prescriptions.
- Receive notifications.
- Use the AI assistant.

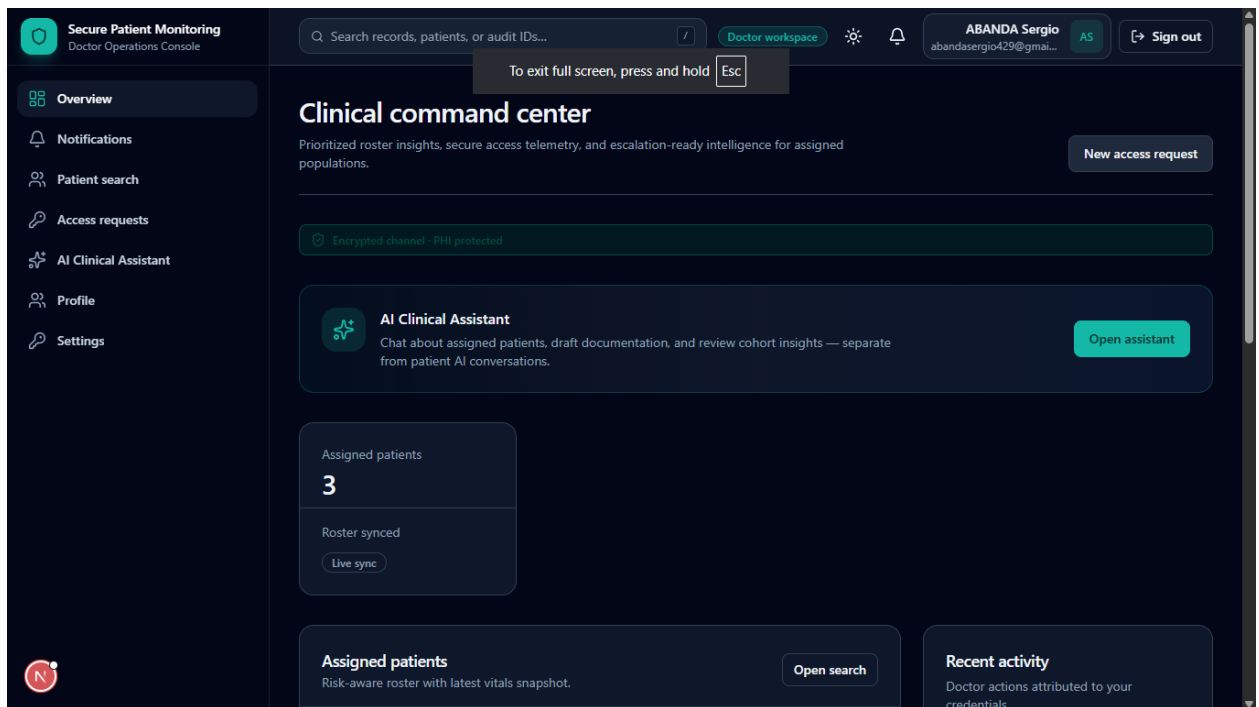


Figure 11: Doctor Dashboard

4.4.5 Hospital Administrator Dashboard

The hospital administrator dashboard provides administrative control over the healthcare institution.

The administrator can:

- Manage doctors.
- Activate or deactivate doctor accounts.
- View doctors associated with the hospital.
- View patients assigned to individual doctors.

- Update hospital information.
- Monitor doctor activities through notifications.

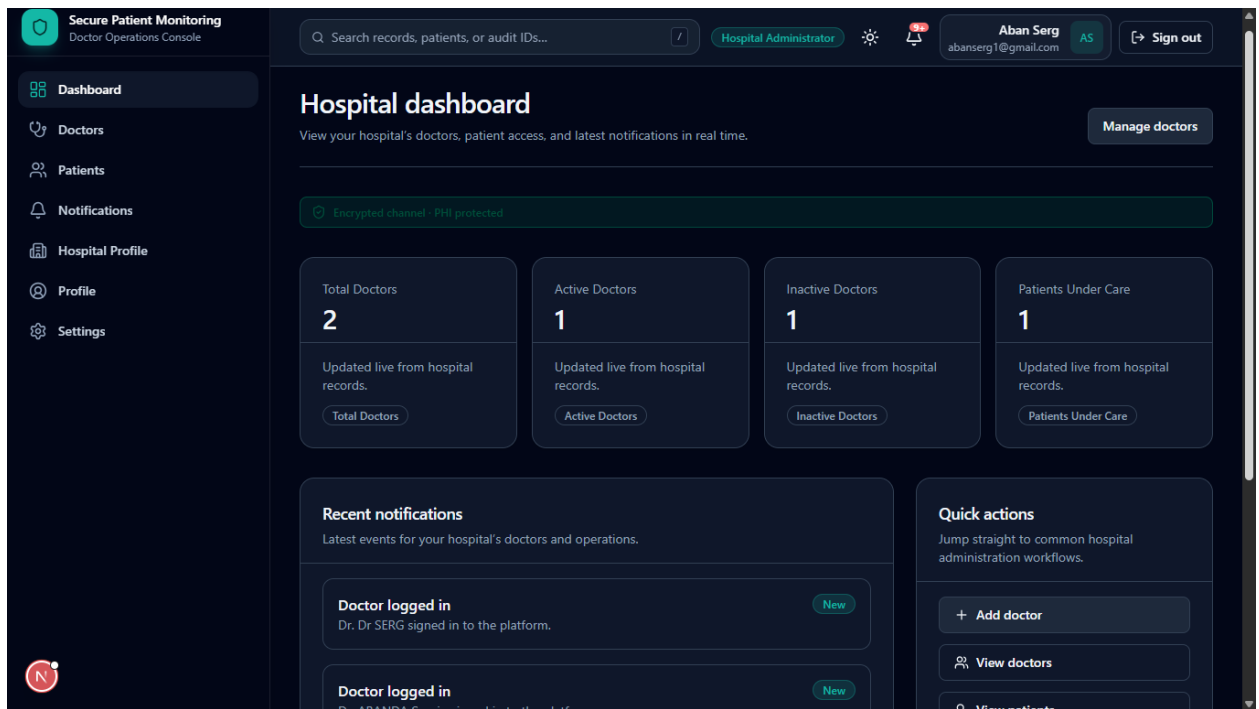


Figure 12: Hospital Administrator Dashboard

4.4.6 Medical Document Management

Patients can securely upload medical documents such as laboratory reports, consultation reports, referral letters, prescriptions, and medical certificates.

Doctors can access these documents only after receiving patient authorization.

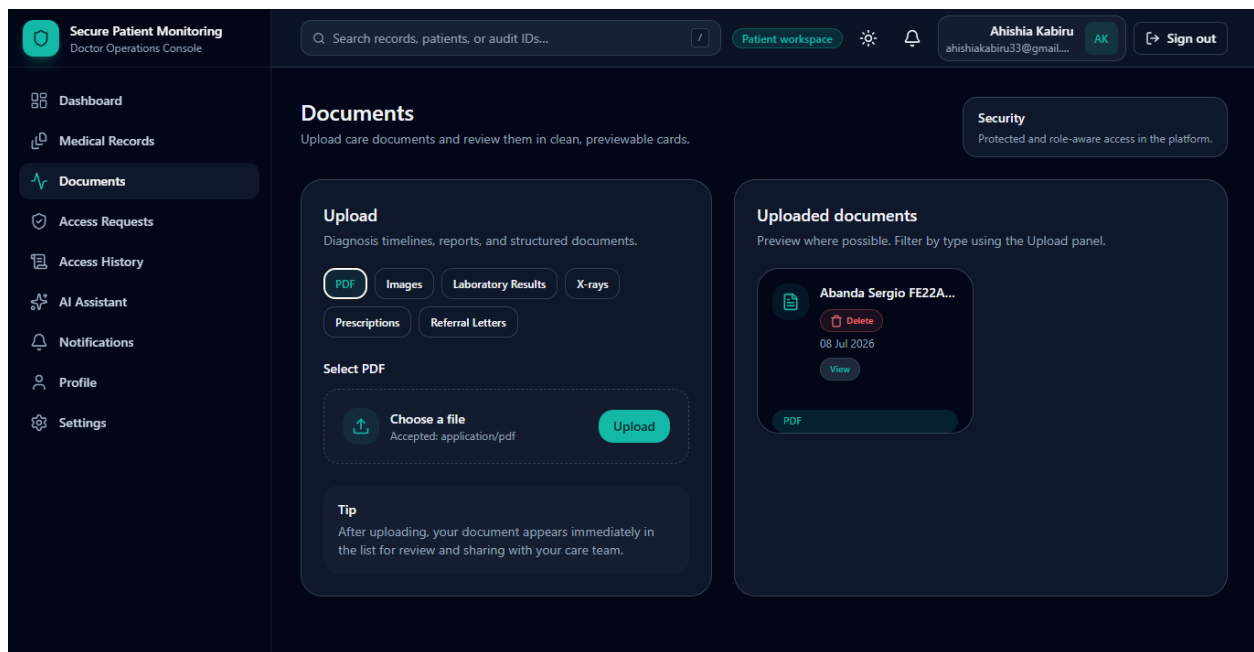
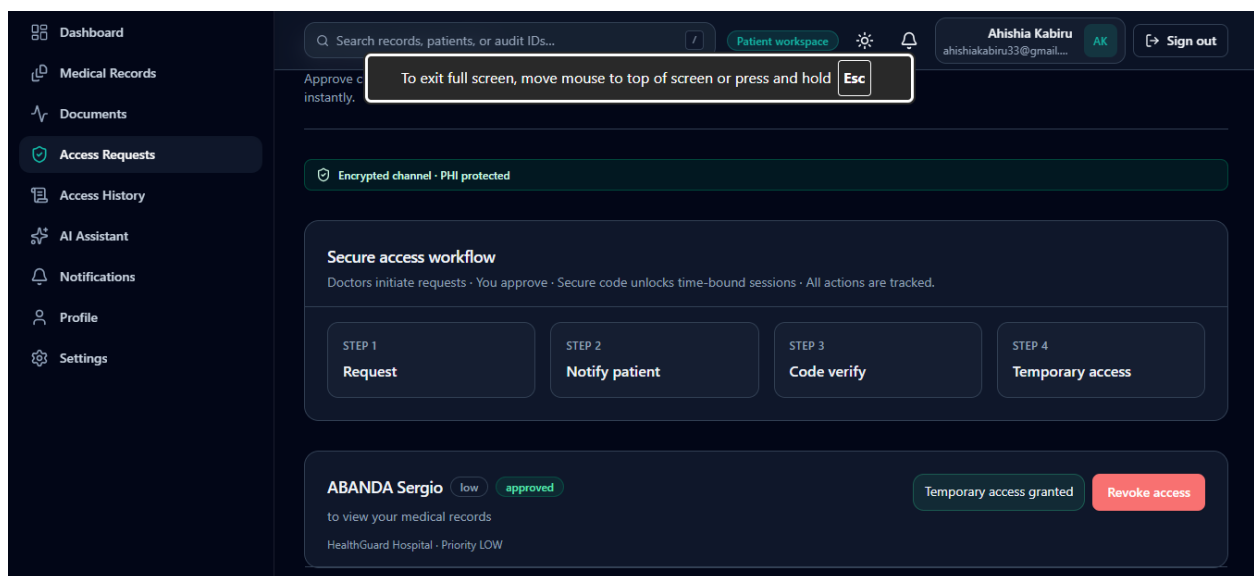


Figure 13: Medical Document Upload

4.4.7 Access Request Management

The access request module enables doctors to request permission before viewing patient medical records.

Patients receive notifications and may approve or reject each request.



Access approval status

Track request progress from submission through temporary privilege activation.

🔒 Encrypted channel - PHI protected



Awaiting patient decision

The patient will approve or reject the request. Notifications are sent to the doctor automatically.

Pending

Open patient monitoring

Return to dashboard

Figure 14: Access Request Module

4.4.8 Notification System

The notification module informs users about important system events such as appointment updates, new prescriptions, access requests, and account activities.

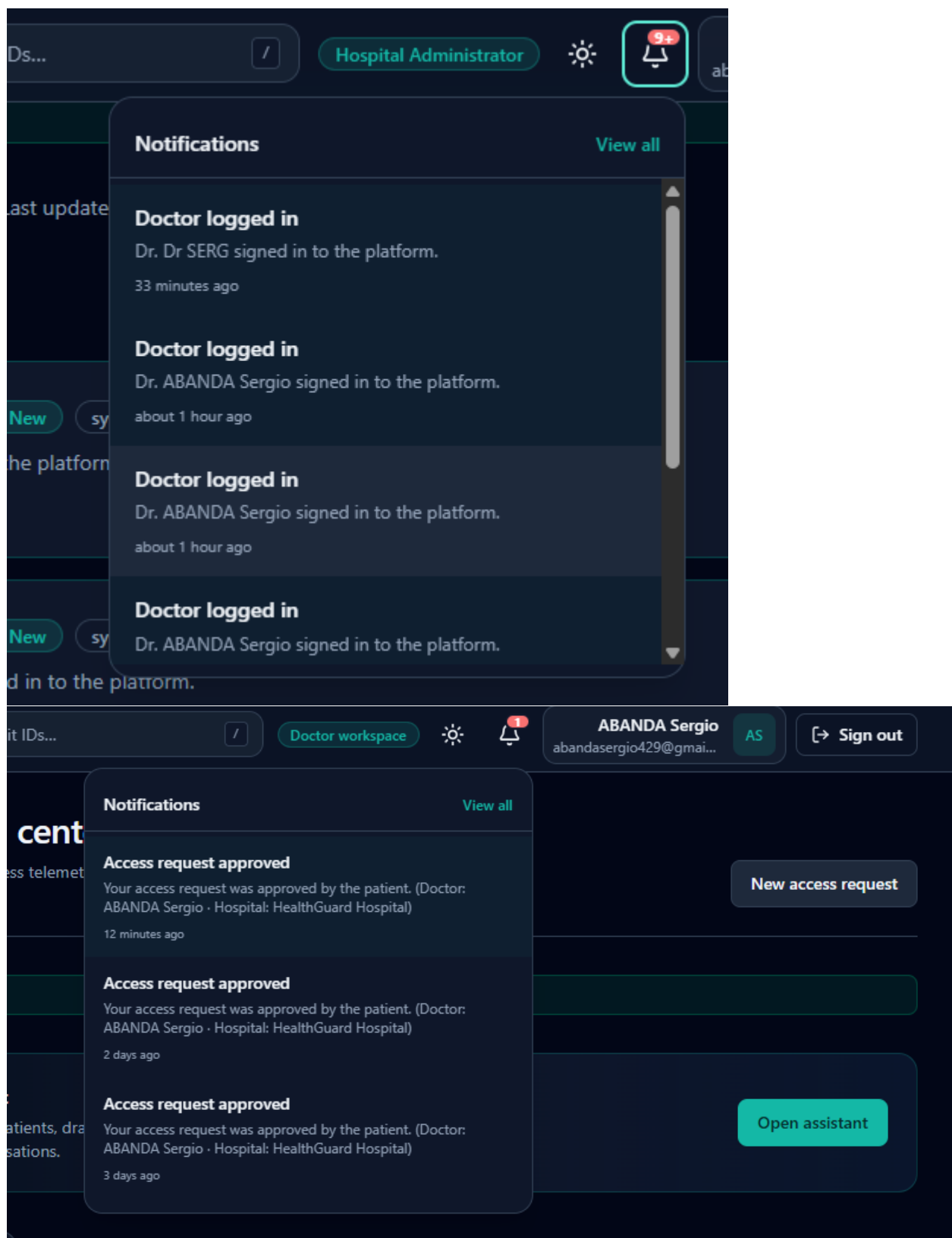


Figure 15: Notification Interface

4.4.9 AI Healthcare Assistant

The AI assistant summarizes uploaded medical documents and answers healthcare-related questions.

To promote responsible AI usage, the assistant is restricted to healthcare-related discussions and encourages users to seek professional medical advice when appropriate.

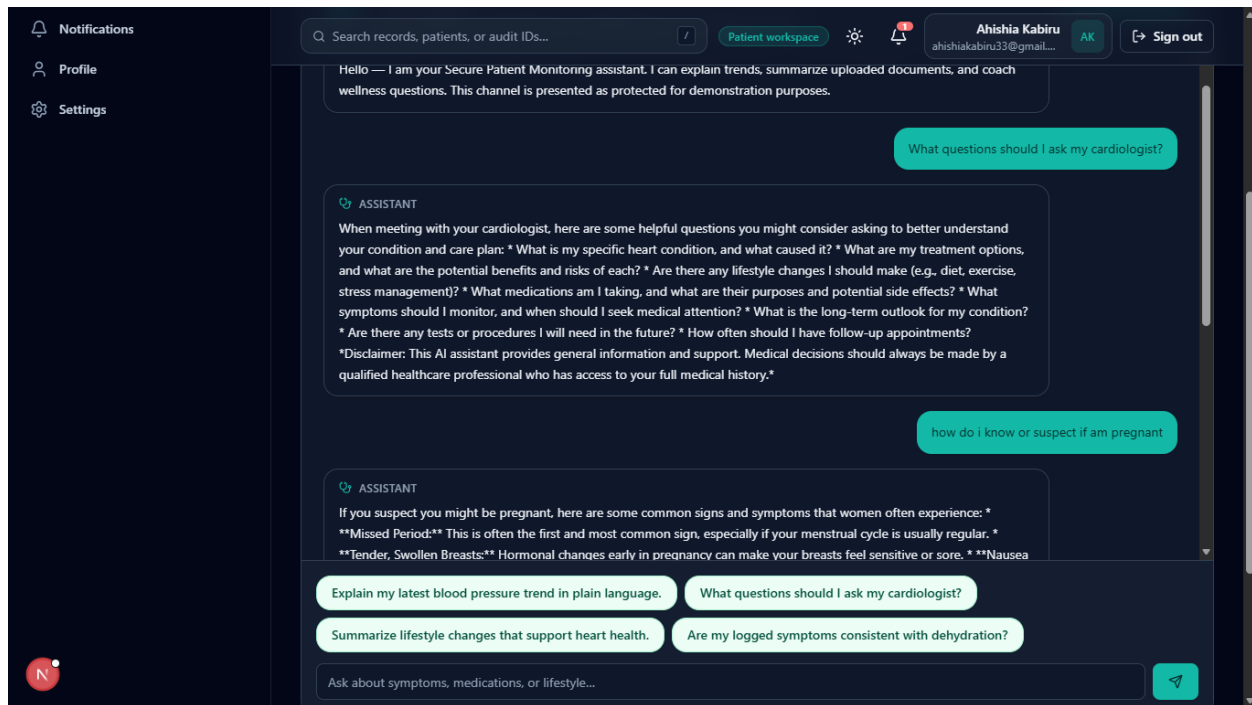


Figure 16: AI Healthcare Assistant

4.5 Evaluation of the Solution

The completed Patient Monitoring System was evaluated by comparing the implemented functionalities with the objectives established in Chapter One.

The evaluation shows that the system successfully achieved its primary objective of providing a secure web-based platform for managing patient medical information.

The implemented role-based authentication mechanism ensures that patients, doctors, and hospital administrators access only the resources authorized for their respective roles. The patient-controlled access request mechanism enhances privacy by ensuring that doctors can only access medical records after receiving explicit patient approval.

The appointment scheduling and prescription management modules improve collaboration between patients and healthcare professionals, while the notification system facilitates timely communication among users.

The integration of Google Gemini Artificial Intelligence further enhances the system by summarizing medical documents, explaining medical reports, and responding exclusively to healthcare-related questions. These capabilities improve the accessibility and understanding of healthcare information without replacing professional medical judgment.

Compared with traditional paper-based patient record management systems, the developed application offers improved security, accessibility, efficiency, and information sharing. The system therefore satisfies the objectives of the study and demonstrates the practical application of modern web technologies and Artificial Intelligence in healthcare information management.

4.6 Partial Conclusion

This chapter presented the implementation of the proposed Patient Monitoring System and demonstrated the major functionalities developed during the project. The tools and technologies used in the implementation were discussed, and the principal interfaces of the system were presented and explained. The evaluation confirmed that the developed application satisfies the objectives of the study by providing a secure, efficient, and user-friendly platform for managing patient medical information. The next chapter concludes the study by summarizing the major findings, highlighting the contributions of the project, discussing its limitations, and proposing recommendations for future improvements.

CHAPTER FIVE: CONCLUSION AND FURTHER WORKS

5.1 Summary of Findings

This project successfully achieved its objective of designing and implementing a secure web-based Patient Monitoring System that improves the management and sharing of patient medical information. The developed system provides separate interfaces for patients, doctors, and hospital administrators, enabling each user to perform tasks according to their assigned roles and privileges.

The system allows patients to securely upload and manage medical documents, schedule appointments, receive prescriptions, approve or reject doctor access requests, and communicate with an Artificial Intelligence healthcare assistant. Doctors can request access to patient records, review authorized medical documents, manage appointments, issue prescriptions, and utilize AI to summarize medical documents and provide healthcare-related assistance. Hospital administrators are responsible for managing doctors within their hospitals, updating hospital information, and monitoring administrative activities through notifications.

The integration of role-based access control, patient-controlled authorization, appointment management, prescription management, notification services, and Artificial Intelligence demonstrates that modern web technologies can significantly improve healthcare information management while maintaining data security and patient privacy.

5.2 Contribution to Engineering and Technology

This project contributes to the field of Software Engineering and healthcare technology by providing a secure, scalable, and user-friendly web-based Patient Monitoring System that addresses several limitations of traditional healthcare information management.

Unlike conventional paper-based record systems, the proposed solution enables secure digital management of patient medical documents while allowing patients to maintain control over who can access their healthcare information. The implementation of role-based access control and patient-approved access requests enhances data confidentiality and privacy.

The integration of Artificial Intelligence using Google Gemini further distinguishes the system from many traditional healthcare applications. The AI assistant assists users by summarizing medical documents, explaining healthcare information in simple language, and responding exclusively to healthcare-related questions, thereby improving the accessibility and understanding of medical information.

Furthermore, the system demonstrates the practical application of modern software engineering principles, including modular software architecture, RESTful APIs, responsive web design, secure authentication, database management using Prisma ORM and PostgreSQL, and Artificial Intelligence integration within a real-world healthcare application.

5.3 Recommendations

Based on the findings of this study, the following recommendations are proposed:

- Healthcare institutions should adopt secure digital platforms to replace paper-based medical record management systems in order to improve efficiency and accessibility.
- Future implementations should strengthen interoperability by integrating the system with existing Hospital Information Systems (HIS) and Electronic Health Record (EHR) platforms.
- Continuous security assessments should be performed to protect sensitive patient information against emerging cyber security threats.
- Healthcare professionals should receive adequate training to ensure effective adoption and utilization of digital healthcare systems.
- Artificial Intelligence should continue to be used as a decision-support tool while ensuring that diagnosis and treatment decisions remain the responsibility of qualified healthcare professionals.

5.4 Difficulties Encountered

Several challenges were encountered during the development of this project.

The integration of multiple technologies, including Next.js, Express.js, Prisma ORM, PostgreSQL, and Google Gemini AI, required extensive research and careful configuration to ensure compatibility between the frontend and backend components.

Database design and management presented additional challenges, particularly in establishing secure relationships among hospitals, doctors, patients, appointments, prescriptions, and access requests while maintaining data consistency and integrity.

Implementing secure role-based authentication and patient-controlled authorization required careful planning to ensure that users could access only the information permitted by their assigned roles.

The integration of Artificial Intelligence also introduced challenges related to API configuration, prompt engineering, and restricting AI responses to healthcare-related topics while preventing inappropriate responses outside the intended domain.

Finally, ensuring a responsive and user-friendly interface across desktop computers, tablets, and mobile devices required continuous testing and interface refinement throughout the development process.

5.5 Further Works

Although the developed Patient Monitoring System successfully satisfies the objectives of this study, several improvements can be implemented in future versions of the application.

Future work may include integrating wearable medical devices and Internet of Things (IoT) technologies to support real-time patient monitoring and automatic collection of physiological data such as heart rate, blood pressure, oxygen saturation, and body temperature.

The system may also be extended to support video consultation and telemedicine services, enabling remote communication between patients and healthcare professionals.

Additional Artificial Intelligence capabilities, such as medical image analysis, predictive health analytics, and clinical decision-support systems, may further improve healthcare service delivery.

Future versions could also incorporate interoperability standards such as HL7 FHIR to facilitate secure exchange of healthcare information with other hospital management systems and national Electronic Health Record platforms.

Finally, deploying the application within multiple healthcare institutions and evaluating its performance under real hospital conditions would provide valuable insights for further optimization and large-scale adoption.

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Appendices

Figure A.1: **Forgot Password – Email Entry Page**

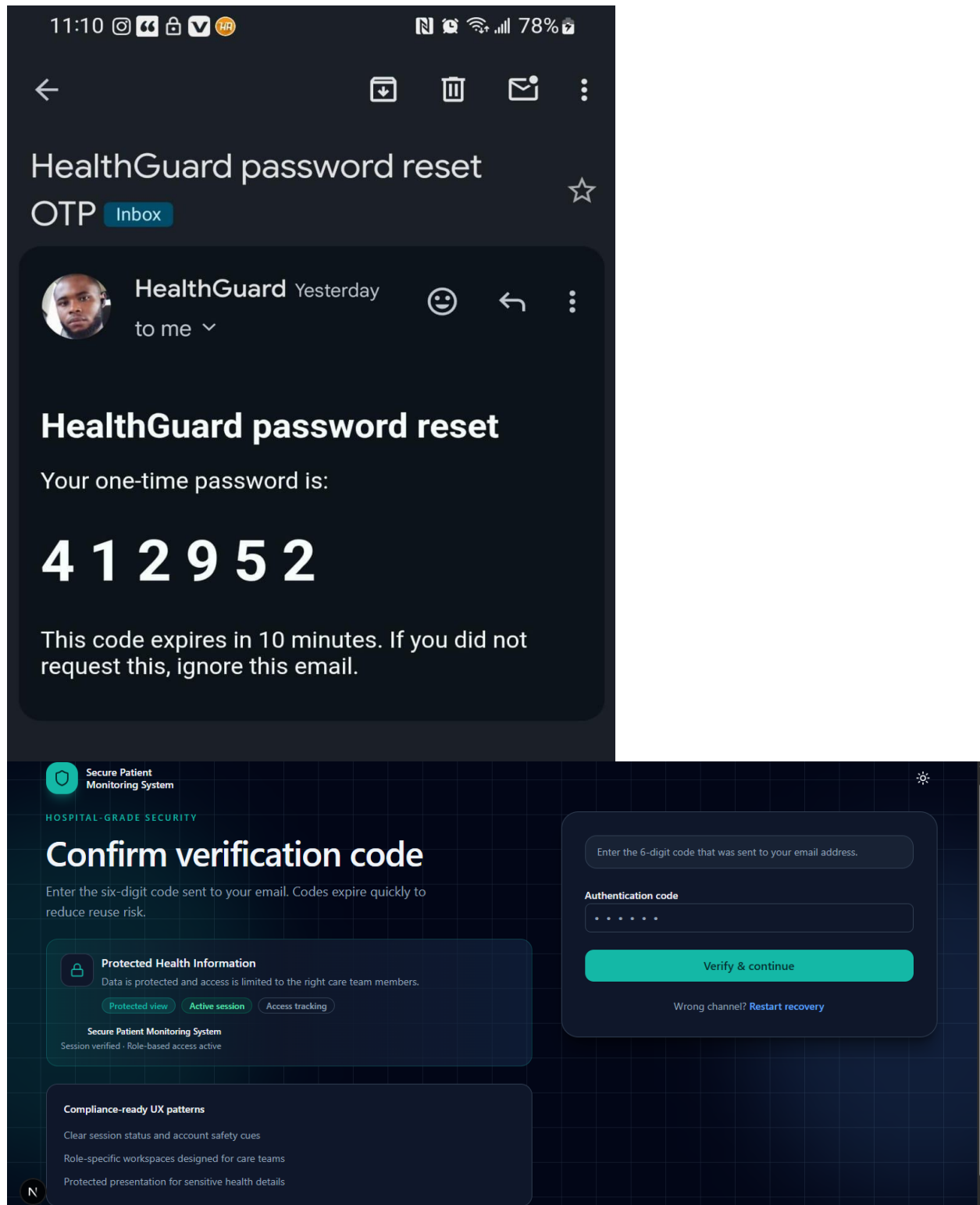


Figure A2: Patient setting interface

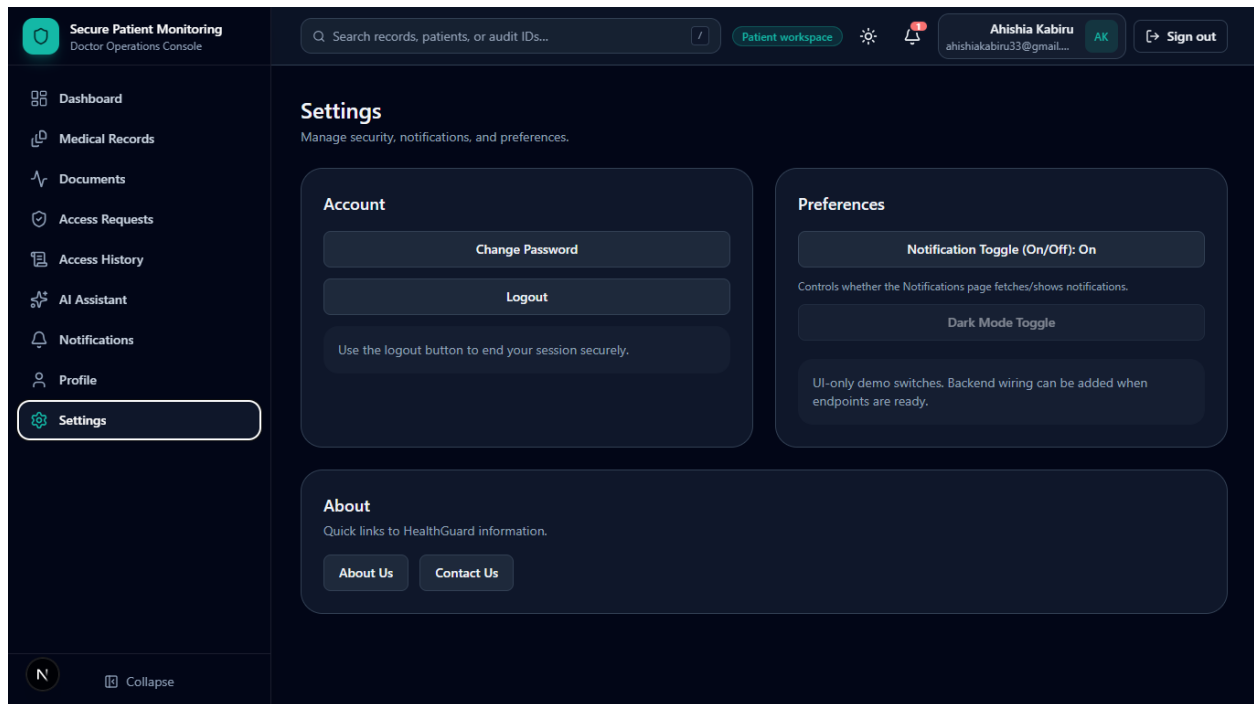


Figure A3: Neon database view

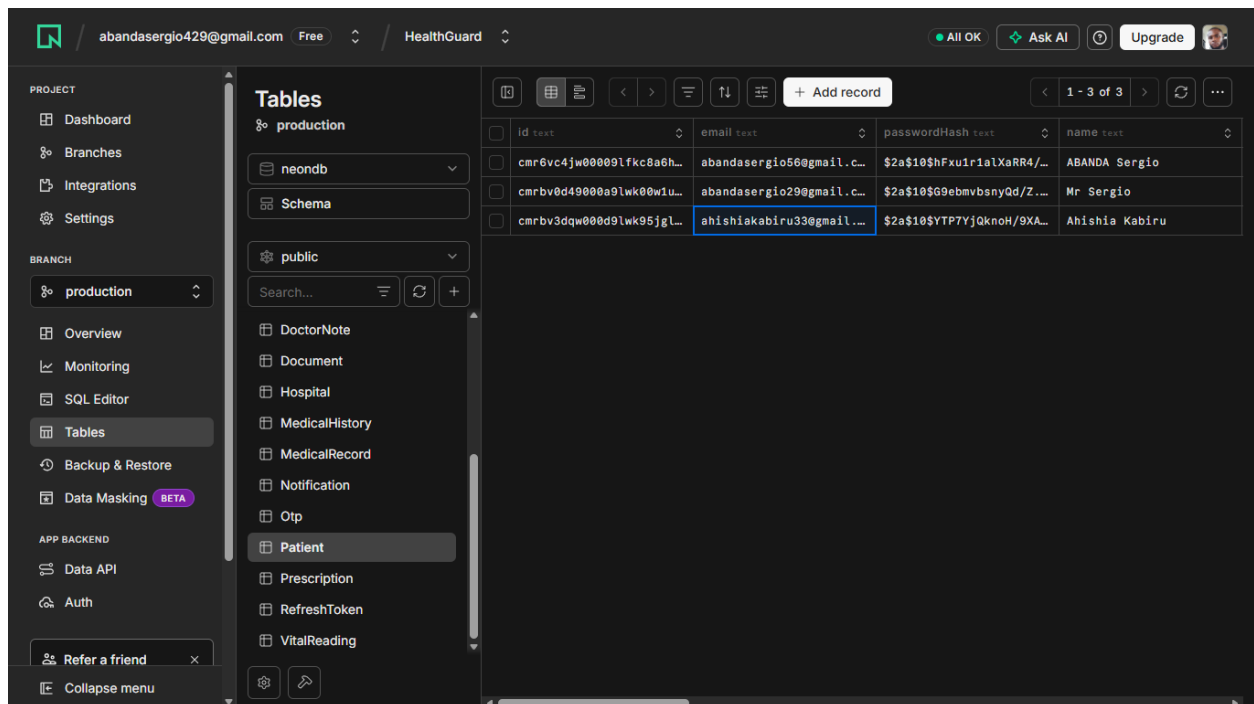


Figure A4: Neon database view

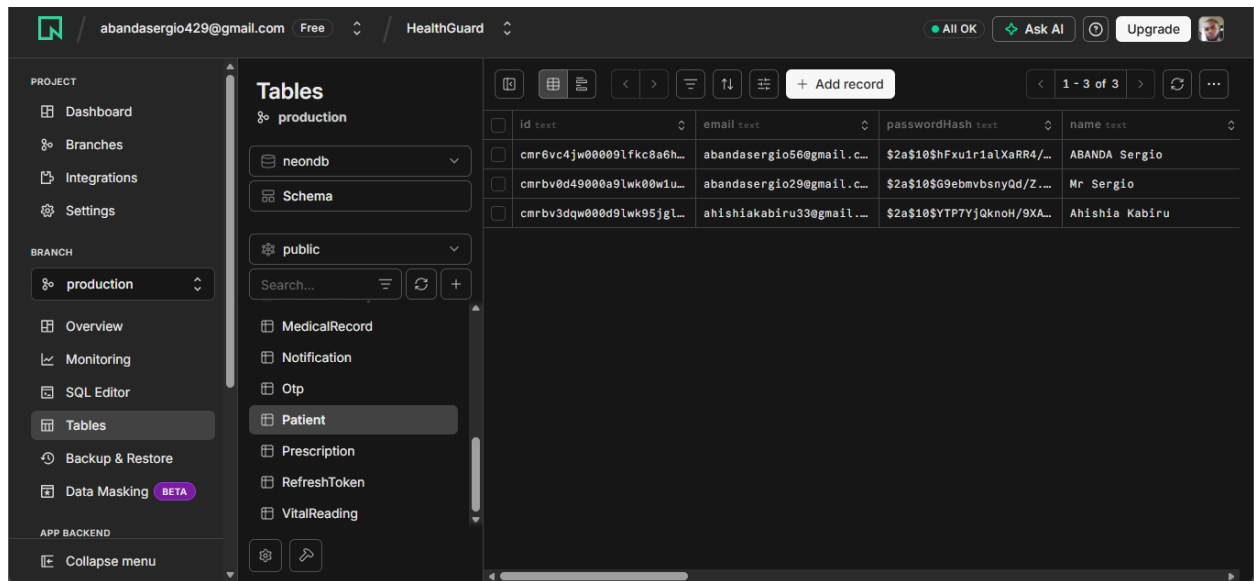


Figure A5: Github Repository of the project

